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A code designed for Geophysics | Seismology | Volcanology

MuRAT

Multi-Resolution seismic Attenuation Tomography

Version 4.0

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MuRAT in Brief

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MuRAT is a Matlab Package for Seismic Attenuation, Scattering and Absorption Tomography using Body and Coda Waves. The quickest way to implement MuRAT is to read his online readme and follow step-by-step his wiki. This PDF document is the complete manual for MuRAT.

Features

What it does: *MuRAT* is a software package for measuring seismic attenuation, scattering, and absorption from passive and active data and model 3D variations of these parameters in space.

The first MuRAT: *MuRAT1.0* was developed by Luca De Siena during his PhD at the INGV-Osservatorio Vesuviano, Italy, and ***published in 2014*** while he was research assistant at the Westfälisches Wilhelms Universität, Münster. Three sample papers published using this code are: ***De Siena et et al. 2017, SR***, ***Prudencio et al. 2020***, and ***Sketsiou et al. 2021***;

In 2D: ***MuRAT2D*** is the result of the activity of the ***Volcano Earth Imaging group***. It produces 2D seismic scattering and absorption maps at multiple frequencies and is suited for largely undetermined problems. Three example papers published using this code are: ***De Siena et et al. 2017, GRL***, ***Napolitano et al. 2020***, and ***Sketsiou et al. 2020***;

In 3D: ***MuRAT3.0 (or MuRAT3)***, now discontinued, has been the standard for measuring total attenuation, scattering and absorption in 3D from 2021. Reference papers are ***Reiss et al. 2022***, ***Gabrielli et al. 2023***, ***Talone et al. 2023***, ***Napolitano et al. 2025***, and ***Talone et al. 2025***.

MuRAT4 : This guide describes the latest implementation of the code with comparison to MuRAT3.

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Chapter 1

Introduction

This document describes how to use the Matlab software package MuRAT to perform *seismic attenuation, scattering and absorption tomography*. This is not a code for imaging *seismic velocity* with travel times or *phase/group velocities* using surface waves. To understand the difference, consider the following seismogram produced by a volcano-tectonic earthquake:

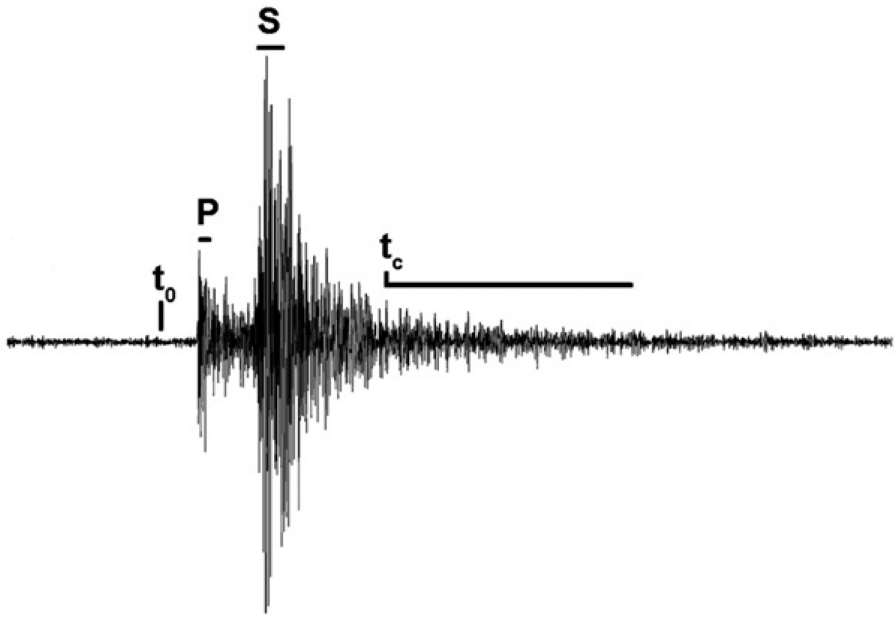


Figure 1.1: A seismogram. MuRAT measures the heterogeneous loss of seismic energy using different seismic attributes

We are used to picking phases on it - for example, the P-wave arrival. We measure velocity changes over different paths by modelling the source-station path followed by the corresponding wave packet with a *ray-tracing* approach. Travel-time tomography requires a *non-linear inversion*, where we must jointly update the paths followed by the waves, the corresponding velocity model, and seismic locations to find a reliable solution. Multiple codes do this: our favourites are LOTOS and FMTOMO (all

underlined names in the manual are links to the related internet sites). Other codes allow the user to measure and image body-wave and surface wave velocity in space (from earthquake and noise) (interfaced with python, like Seislib) and full waveform inversions (like SPECFEM3D). Combining MuRAT with velocity tomography is highly recommended for complete imaging and modelling of the area under study.

MuRAT deals with the energy loss seismic waves suffer while propagating through the heterogeneous Earth. It fills the gap for open-access tools to measure attenuation, separate the processes that lead to it, and model them in space. To our knowledge, the first mention of an *attenuation tomography* was given by Ho-Liu et al. (1988) in the Coso Valley. More than 30 years afterwards, this technique was a standard with global, regional and local applications. Yet, different mechanisms attenuate seismic waves while they propagate. While codes to model total attenuation in space exist, no open-access code could measure attenuation, separate the mechanisms that produce it and model their variations in space with seismic inversion using as input only seismograms downloaded from the data servers.

We define here the mechanisms causing seismic attenuation:

1. Seismic energy progressively spreads over larger volumes while propagating far from the source. The energy thus decreases with distance due to *geometrical spreading*. This mechanism is independent of frequency for a given phase. However, depending on frequency, often more than one phase is included in a *window of time* measured on a seismogram - in Fig. 1.1, it is challenging to distinguish between S-waves and surface waves produced right after them.

2. Seismic energy is lost into the heterogeneous Earth depending on the ratio between the wavelength (λ) and the dimension of the heterogeneity (a). The energy thus decreases with distance due to *scattering losses*. This mechanism depends on the frequency and comprises complex physics, as those described by multi-pathing and diffractions. Reflection and refraction can be characterised as scattering processes, although they only affect coherent waves.

3. Seismic energy interacts with the heterogeneous Earth losing heat for each cycle. The energy thus decreases with distance due to *absorption losses* in the materials they pass through. This mechanism depends on frequency. Quantifying how much energy the Earth absorbs is challenging to, but such a quantification is essential to connect seismology with techniques that measure potential fields and heat losses.

4. The sum of scattering and absorption losses gives *total attenuation*. This is the attenuation experienced by direct wave packets while they propagate through the heterogeneous Earth.

When we talk about seismic attenuation, we generally refer to total attenuation.

MuRAT measures and models total attenuation, scattering attenuation and absorption in space using single seismogram observations. The first three chapters of this documentation offer an overview of the theory underlying attenuation tomography in these three forms.

In chapter 2, we revise the theory behind measurement and modelling of the energy lost by direct waves (**P** and **S**, Fig. 1.1). The method used to model total attenuation in chapter 2 is the *coda normalisation* (CN) method (Aki, 1980; Yoshimoto et al., 1993). In tomography, the method has been introduced by Del Pezzo et al. (2006a). At the core of MuRAT are seminal papers, which have been developed by Del Pezzo et al. (2006b), De Siena et al. (2009) and De Siena et al. (2010) for multi-scale applications. De Siena et al. (2010) benchmark the CN with the standard spectral slope (t^*) method, applied by the global community. For a decade, the CN method has only been applied to volcanoes. The reason is the need for a relevant *coda* (t_c, Fig. 1.1) to normalise direct-wave information. The method has the advantage of removing source and site effects from direct wave measurements but suffers from an inexact description of the process. MuRAT uses the standard description of geometric spreading, r^2, with r being the hypocentral distance. MuRAT4 deals with the problem associated with direct-wave measurements and modelling, using theoretical and computational results obtained throughout the last decade, where the method has been applied to fracture networks (Napolitano et al., 2025) and subduction zones (Feng et al., 2026).

In chapter 3, we revise the theory behind the use of *peak delays* as markers of scattering losses (Saito et al., 2002; Takahashi et al., 2007, 2009; Tripathi et al., 2010; Calvet and Margerin, 2013; Calvet et al., 2013; De Siena et al., 2016; Sato, 2016). Peak delays are a focus of the early-warning community and a quantity important to seismic source modelling in the near field. The standard theory recognises peak delay as a marker of scattering losses in the far field, modelled by the Markov approximation (Saito et al., 2002), in the case scattering is produced only by long wavelength component ($\lambda << a$). However, recent research also suggests a dependence of peak delay on scattering when λ is of the order of propagation distance and correlation length of heterogeneity (Napolitano et al., 2020; Di Martino et al., 2021). Here, phenomena like diffraction and trapping take over (King et al., 2023; Gabrielli et al., 2023). The chapter discusses the standard application, theoretical limitations and future outlooks, like applications across fracture networks at local (Napolitano et al., 2025) and regional (Talone et al., 2023) scales.

In chapter 4, we revise the theory behind the use of *late-time coda attenuation* as marker of seismic absorption (Wegler and Luehr, 2001; Prudencio et al., 2013; Calvet and Margerin, 2013; Del Pezzo et al., 2016; De Siena et al., 2016, 2017a; Del Pezzo et al., 2018; Sketsiou et al., 2020). Its application relies on a preliminary Multiple-Lapse-Time-Window-Analysis (MLTWA - Fehler et al. (1988)). Coda attenuation can be described using single and multiple scattering. MuRAT provides the tools to test the onset of equipartition, a necessary condition for diffusion, where coda attenuation is equivalent to seismic absorption. The embedded sensitivity kernels model coda attenuation in space based on Paasschens equations (Paasschens, 1997) and the approximations of Pacheco and Snieder (2005). The theory is inexact in the case of sharp vertical or lateral boundaries, so much so that diffusion could never onset Morozov (2008). However, in zones where the Moho is thick and for sufficient data, the theory generally produces stable results (Sketsiou et al., 2020). New work has been recently developed to include the effect of sharp boundaries and vertical velocity variations on coda attenuation measurements (Sanborn et al., 2017; Cormier and Sanborn, 2019; Nardoni et al., 2021). When the correct assumptions are used, absorption imaging has demonstrated the ability to image melt and pathways followed by dyke intrusions (Reiss et al., 2022), temperature (Yemane et al., 2025), and gas saturation changes (Talone et al., 2025). MuRAT4 provides the tool to invert coda attenuation as marker of absorption using a diffusive approximation and differential sensitivity kernels.

From chapter 5, this manual describes how to install and use the code, so you can start from there if you already know the theory.

Chapter 5 describes installation procedures, online documentation tools, and data preparation. MuRAT4 and its Readme.md are available on GitHub and have a Wiki explaining the primary steps to run the code and interpret results. We discuss why the code is released in proprietary (Matlab©) language and the corresponding installation procedures. The code accepts only Seismic Analysis Code (SAC) data. SAC is a standard for seismologists and the favourite format when converting miniseed on data servers like IRIS. You may have been processing data with another software. This chapter will thus discuss the conversions necessary to create a readable dataset for MuRAT, especially the tools you can use to assess if the data are in the right format, created directly in Matlab.

Chapter 6 describes the input files. We offer three example applications (Mount St. Helens, US - Toba, Indonesia - Vrancea, Romania). In theory, these are the only files the users need to create once data files are in the correct format.

Chapter 7 describes the output structures and files. The chapter describes all outputs, where

you find them and what they represent.

Chapter 8 describes the MuRAT tests. This is a short description of the tests used to assess results, especially of its figure outputs, including those to assess damping.

Chapter 9 describes rays and kernels outputs. The figure outputs show rays and kernels used for each measurement and discuss the transformations from geographic to cartesian.

Chapter 10 describes the inversion. This chapter explains how to set damping and smoothing parameters using L-curves, and describes the alternative global optimization strategies available in MuRAT4.

1.1 Matlab

MuRAT wants to be an open-access code, yet it also comes in Matlab format: why?

- The first reason is historical. The code has been designed in Matlab, with legacy bash and Fortran codes embedded in it. Passing from Matlab to Python, the obvious choice for open-access interfaces was not straightforward. Passing from Matlab to Julia has been way easier. Evolving through a decade, the Matlab version of MuRAT has reached the format presented here. MuRAT4.0 is the current and maintained Matlab release.

- The second reason is the simplicity of learning. MuRAT has been thought of as a way to introduce students to computational seismology. Matlab is an ideal interface for teaching computational codes to students in Earth Sciences. MuRAT has also been developed with the wider geological community in mind, which until recently was often lacking training in computation. The average geology student understands Matlab better than other codes.

- The third reason is that Python is becoming the standard in seismology, mainly thanks to tools like Obspy, yet this is not so obvious for the rest of Earth Sciences. For example, Matlab is (has been) the favourite language interface for most of the Geodynamics community - see, for instance, Taras Gerya's Introduction to Numerical Geodynamic Modelling. The geodynamics community is now embracing Julia as its natural successor. MuRAT does the same.

Chapter 2

Direct-wave attenuation: Q

Timeline in papers.		
Paper	**Descriptor**	**Where**
(Del Pezzo et al., 2006a)	S-wave CN tomography	Link
(De Siena et al., 2010)	Benchmark with t^* method	Link
(De Siena et al., 2014a)	MuRAT1.0 for CN method published	Link
(De Siena et al., 2014b)	P-wave coda-normalisation tomography	Link
(De Siena et al., 2017b)	CN tomography measuring Qc	Link
(Napolitano et al., 2020)	Feature work with MuRAT2D	Link
(Reiss et al., 2022)	Scattering/absorption with MuRAT3D	Link
(Talone et al., 2023)	Scattering/attenuation with MuRAT3D	Link

2.1 Parameters

Input Parameters for MuRAT4		
Symbol	**Descriptor**	**Where**
t_P or t_S (s)	P-wave or S-wave arrivals	SAC Header
t_l (s)	Length of P- or S-wave window	Set in input file
t_0 (s)	Origin time	SAC Header
t_c (s)	Lapse time from origin time	Set in input file
t_w (s)	Length of coda window	Set in input file

2.2 The coda normalisation method in pills

The method measures direct and coda-wave energies in frequency and divides them to obtain a quantity that only depends on Q, allowing for a linearised inversion.

Pros of measuring direct-wave attenuation using coda-normalised energies

1. Single-station measurements - you can get all the info from a single seismogram.
2. No need for correcting source and site effects.

Cons of measuring direct-wave attenuation using coda-normalised energies

1. Uncertain sensitivity of coda waves to Earth structures - homogeneous coda wave sensitivity assumption is generally unfulfilled. *Tackled by MuRAT estimating the measurable coda-wave energy at the chosen lapse time and testing equipartition.*

2. Correction of radiation pattern - increasing direct wave windows smooth it at the expense of focusing. *Tackled by testing different direct-wave windows to measure energy.*

Figure 2.1 is a visual representation of the method.

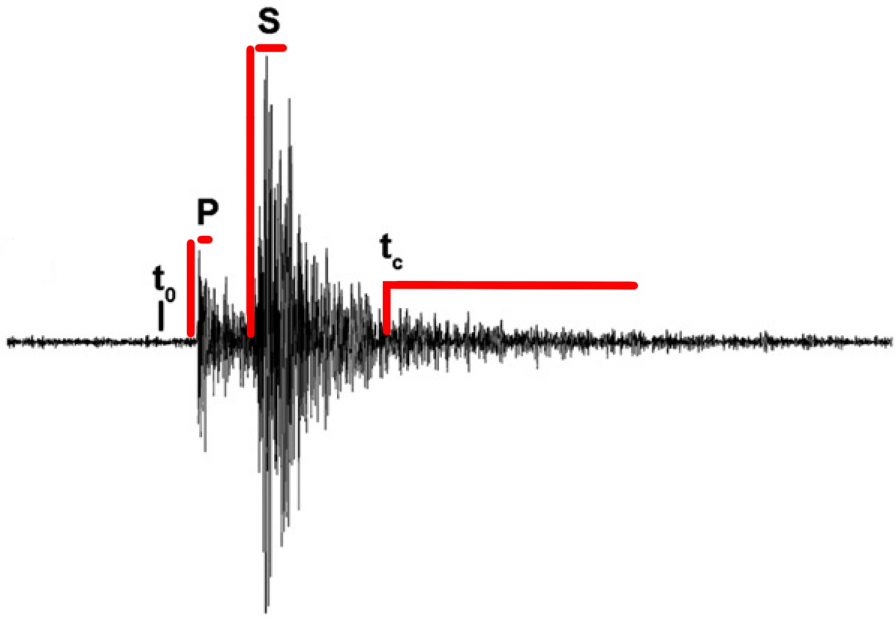


Figure 2.1: The CN method measures the spectral energies of direct and coda waves and inverts for Q. The P and S wave energies, and the normalising coda energies are in red.

2.3 The theory bit

2.3.1 The classical coda-normalization method

The CN method normalises P- and S-wave energy for coda energy (Aki, 1980; Yoshimoto et al., 1993). Seismic wave energies can be modelled as the convolution of the source, path, site, and instrument functions so that the spectral energy of direct waves (E) is:

$$E(f,r) = R_{\theta\phi} S_S(f) r^{-\gamma} I(f) G(f,\Psi) \exp\left(-\frac{2\pi f}{Q(f)v} r\right) \tag{2.1}$$

where ***f*** is the frequency, r is the source-receiver distance, t_c is the coda-wave central lapse time from the origin time of the event (t_0), $R_{\theta\phi}$ is the source radiation pattern (θ and ϕ the azimuth and take-off angle for a source-receiver ray), $S_S(f)$ is the source function, γ the geometrical spreading exponent, $I(f)$ the known instrumental response, $G(f,\Psi)$ is the site amplification factor (with Ψ being the incident angle of the ray at the station), Q is the direct-wave quality factor and v the average velocity in the medium (Yoshimoto et al., 1993).

The spectral energy of coda waves (E_c) can be written as:

$$E_c(f,t_c) = S_S(f) P(f,t_c) G(f) I(f). \tag{2.2}$$

The attenuation of coda waves can then be expressed as (Aki and Chouet, 1975):

$$P(f,t_c) \simeq t_c^{-n} \exp\left(-2\pi f Q_c^{-1} t_c\right) \tag{2.3}$$

In this equation, n is the envelope spectral decay, Q_c is the coda quality factor, and $A_c(f,t_c)$ does not include the effect of the source radiation pattern. We will discuss more about this term in Chapter 4.

Eq. 2.1 can be divided by Eq. 2.2 to normalise the energy of the spectral energy of direct waves using the spectral energy of coda waves:

$$\frac{E(f,r)}{R_{\theta\phi} E_c(f,t_c)} = r^{-\gamma} \frac{G(f,\Psi)}{G(f)} \exp\left(-\frac{2\pi f}{Q(f)v} r\right) \frac{1}{P(f,t_c)} \tag{2.4}$$

where the source function $S_S(f)$ and the instrumental response $I(f)$ contributions disappear. By measuring direct-wave energy over a window of sufficient length (t_l), we smooth the azimuthal contribution of the radiation pattern (De Siena et al., 2009, 2010). If the length of the direct-wave window is chosen appropriately, the contribution of the source radiation pattern $R_{\theta\phi}$ is negligible (De Siena et al., 2009) so that $R_{\theta\phi} = 1$ and $\frac{G(f,\Psi)}{G(f)} = 1$. Since the early coda consists of randomly scattered waves, the coherency and the source radiation pattern are eventually lost (Takemura et al., 2009). De Siena et al. (2010) demonstrate that a coda window of 2 s is sufficient to make the radiation pattern quasi-isotropic in a volcanic caldera. Still, the window length must be carefully chosen at each frequency to avoid the near-receiver onset of surface waves (Gabrielli et al., 2020).

At fixed frequency band and starting lapse-time for coda windows, $P(f, t_c)$ is assumed to be constant in the standard coda-normalisation method (Del Pezzo et al., 2006a; Sato et al., 2012). Taking the logarithm, Eq. 2.4 becomes:

$$\ln\left[\frac{E(f,r)}{E_c(f,t_c)}\right] = -\ln P(f,t_c) - \gamma \ln(r) - \frac{2\pi f}{Q(f)v} r, \tag{2.5}$$

a linear equation solved for three unknown ($\ln P(f,t_c)$, γ, Q) once the hypocentral distance is known and the frequency is set. If Q_c is known for each source station pair, one can pre-estimate $P(f,t_c)$ using equation 2.3, as done by De Siena et al. (2017b):

$$\frac{1}{2\pi f}\ln\frac{E(f,r)}{E_c(f,t_c)} + \frac{1}{2\pi f}\ln P(f,t_c) = -\frac{\gamma \ln(r)}{2\pi f} - Q(f)^{-1} t, \tag{2.6}$$

where t is the travel time of the corresponding direct phase. The system can be solved with a linear inversion for γ and $Q(f)^{-1}$. This is the method implemented in MuRAT3.0.

2.3.2 Applicability

For the applicability of the CN method, one must measure direct-wave energy. This seems simple enough, but it can become problematic depending on frequency, heterogeneity and source-station distance. To check the validity, one must always look at the relationship between direct-wave energy and epicentral distance. Is there a dependence? If not, near field or strong scattering might be prevalent. Does the relationship follow standard geometrical spreading, like those for direct wave energy? How important is the surface wave component? A positive is that generally, where you cannot measure *direct-wave energy*, you can use codas, i.e., diffusion onsets pretty quickly.

2.4 MuRAT3 workflow

MuRAT3 worked in analogy to De Siena et al. (2010), De Siena et al. (2014a) and Sketsiou et al. (2021) when mapping total attenuation. These authors show that one must check the relationship between travel time and measured energy ratio. In Figures 2.2, we show an example at Mount St. Helens volcanoes, where sources and stations can be either on or inside the volcanic edifice and near surroundings (red dots) or far from them (cyan dots). The expected energy decrease with hypocentral distance is unfulfilled at small distances, where surface and trapped waves dramatically increase coda

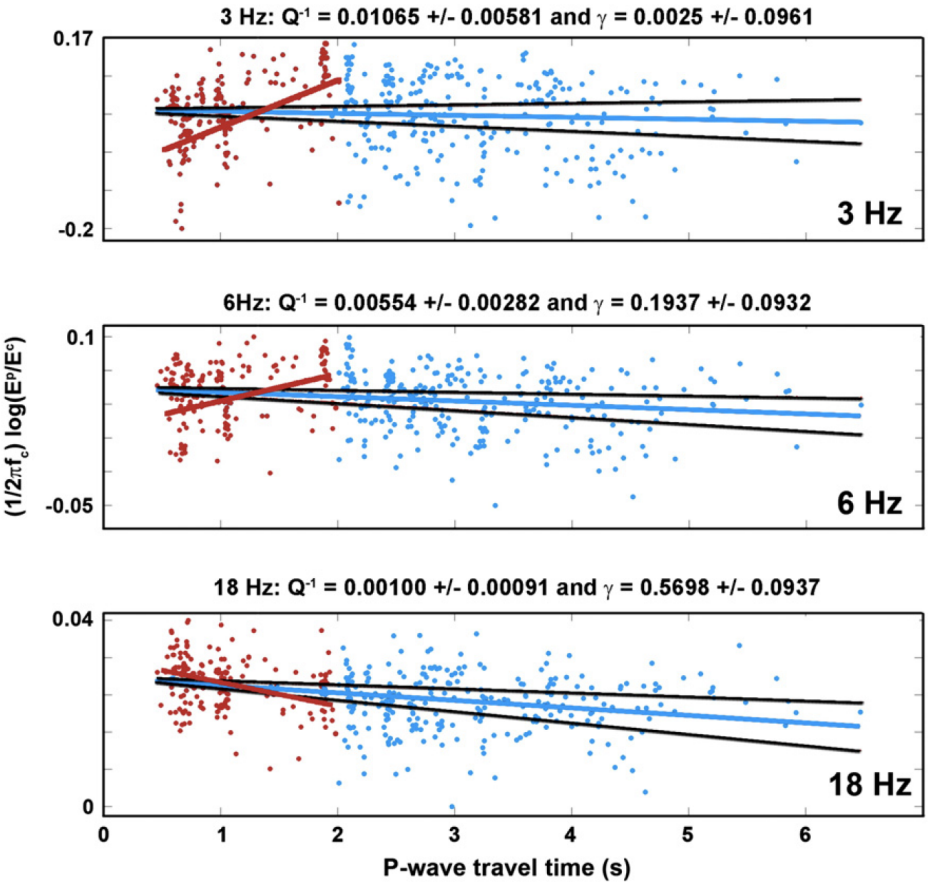


Figure 2.2: A plot of the logarithm of the P-wave direct-to-coda energy ratios (red and cyan dots, computed at MSH in three frequency bands) versus the P-wave travel times at Mount St. Helens volcano. The red lines show the linear fit obtained using only data between 0 and 2 s travel times, affected by surface waves inside the volcanic edifice. The cyan lines represent the least squares fit for the entire dataset. The black lines are the maximum uncertainties. The geometrical spreading and the average Q obtained from the inversion are shown above each plot. Notice the progressive change with increasing frequency. From De Siena et al. (2014a).

energy.

The energy in the coda can be back-traced to mark the position and intensity of the corresponding scatterers (De Siena et al., 2014b), using scattering tomography (also known as Nishigami (1991) technique). The method can still be applied, but fixing the geometrical spreading will trade-off measurements at low frequencies. Sketsiou et al. (2020) discusses the corresponding trade-offs in volcanic,

fault and regional settings. In Fig. 2.3, we show the same plot for the normalised S-wave energy at Mount Vesuvius, where all sources and stations are inside the more heterogeneous volcanic edifice and feeding system domain. Here, only data at 18 Hz can be used. Following De Siena et al. (2014a), measurements above (below) the logarithmic, linear trend show high (low) attenuation. This procedure is inverted by this version of MuRAT using a ray bending approach (Block, 1991; De Siena et al., 2010), discussed computationally in Chapter 10.1.

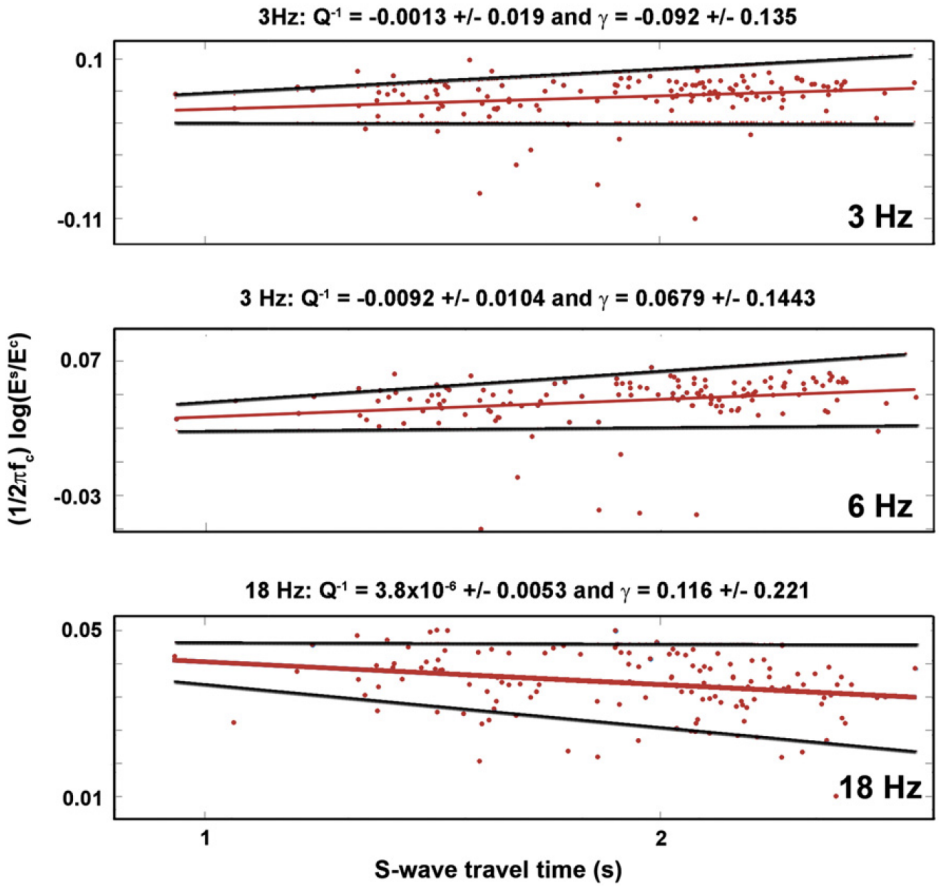


Figure 2.3: Same as Fig. 2.2 for Mount Vesuvius, where earthquakes and stations are all comprised under the volcanic edifice. Notice the negative Q at low frequencies. From De Siena et al. (2014a).

2.4.1 Diffusion assumption on coda or the new CN method

For the implementation in MuRAT4, we rewrite eq. 2.1 for a fully-coherent S wave, thus one with a typical geometrical exponent of $\gamma = 2$, assuming we can remove dependency on radiation pattern and Ψ. In the same framework used for the diffusive coda, the time-integrated energy density of the direct wave is (Sato et al., 2012):

$$E(f,r) = \frac{S_S(f)I(f)G(f)}{4\pi v r^2} \exp\left(-\frac{2\pi f}{Q(f)v} r\right). \tag{2.7}$$

Also eq. 2.3 did not assume a scattering regime for the coda window. However, as we will see in Chapter 4, in MuRAT the coda window is chosen in order to consider diffusive waves (Calvet and Margerin, 2013). Thus, we can rewrite this equation with the additional dependence on the diffusion constant D and absorption Q_I^{-1} as (Sato et al., 2012):

$$E_D(f,t) = \frac{S_S(f)G(f)I(f)}{(4\pi D t)^{1.5}} \exp\left(-\frac{r^2}{4Dt} - 2\pi f Q_I^{-1} t\right). \tag{2.8}$$

Absorption will be measured using coda wave attenuation Q_C^{-1}, its analogue in the diffusive regime (Calvet and Margerin, 2013), i.e., $Q_I^{-1} = Q_C^{-1}$.

However, we measure coda-wave energy across a high-lapse-time window, not at a single lapse time (see Chapter 4). MuRAT integrates the coda energy between the starting- and end-time of the window, t_s and $t_e = t_s + t_w$, and rescales it to the length of the direct-wave window t_l, so that coda and direct energies are integrated over windows of equal length. The measured coda energy is thus t_l times the mean diffusive energy in the window:

$$E_c(f,t_s,t_e) = \frac{t_l}{t_e - t_s} \int_{t_s}^{t_e} E_D(f,t)\, dt = \frac{S_S(f)G(f)I(f)\, t_l}{(4\pi D)^{1.5}}\, \overline{P}_D(f,r), \tag{2.9}$$

where

$$\overline{P}_D(f,r) = \frac{1}{t_e - t_s} \int_{t_s}^{t_e} t^{-1.5} \exp\left(-\frac{r^2}{4Dt} - 2\pi f Q_C^{-1} t\right) dt \tag{2.10}$$

is always positive. Note that the difference between the diffusive energies at t_e and t_s cannot replace this integral: it is negative whenever the diffusive energy decays in the window, which is the case for

most rays at the lapse times used in MuRAT. The new CN equation thus becomes:

$$\frac{E(f,r)}{E_c(f,t_s,t_e)} = \frac{(4\pi D)^{1.5}}{4\pi v\, t_l\, r^2} \exp\left(-\frac{2\pi f}{Q(f)v} r\right) \overline{P}_D(f,r)^{-1} \tag{2.11}$$

Taking the logarithm, moving the geometrical spreading of the right-hand side toward the left, and dividing by $2\pi f$, this becomes:

$$\frac{1}{2\pi f} \ln\left(\frac{E(f,r)}{E_c(f,t_s,t_e)} r^2\right) = -\frac{r}{Q(f)v} + \frac{1.5}{2\pi f} \ln(4\pi D) - \frac{1}{2\pi f} \ln \overline{P}_D(f,r) - \frac{1}{2\pi f} \ln(4\pi v\, t_l) \tag{2.12}$$

The diffusion constant depends on both albedo (B_0) and inverse extinction length (L_e^{-1}), (measured independently, see Chapter 4) (Wu and Aki, 1988; Wegler, 2003):

$$D = \frac{v}{3B_0 L_e^{-1}}. \tag{2.13}$$

We can rewrite the left hand side as our data vector $d = \frac{1}{2\pi f} \ln\left(\frac{E(f,r)}{E_c(f,t_s,t_e)} r^2\right)$, while the last three terms on the right hand side are a coda constant c_{coda} defined for each source-station pair using coda analysis:

$$d = -Q(f)^{-1} t + c_{coda}(D, r, Q_C^{-1}), \tag{2.14}$$

where we defined the travel time $t = r/v$ and c_{coda} as:

$$c_{coda} = \frac{1}{2\pi f} \left[1.5 \ln(4\pi D) - \ln \overline{P}_D(f,r) - \ln(4\pi v\, t_l)\right]. \tag{2.15}$$

In MuRAT, $\overline{P}_D$ is computed numerically with the trapezoidal rule, and v is the average S-wave velocity. Equation 2.14 shows that our data d have sensitivity to both total attenuation and a complex combination of B_0 and L_e^{-1}. Since c_{coda} changes with distance, and distance correlates with travel time, a single constant for all rays would bias Q^{-1}. MuRAT4 thus removes c_{coda} ray by ray and offers two options (**Murat.input.diffusionConstant**):

- **'Theoretical'**: D is fixed by eq. 2.13, using the input B_0 and L_e^{-1}, and the average Q^{-1} is the

least-squares solution of eq. 2.14.

- **'Inverted'**: for each trial D (a logarithmic grid between 10^{-3} and 10^{3} km^2/s, refined locally), the average Q^{-1} is obtained by least squares, and MuRAT keeps the D that minimises the misfit.

In the second case, MuRAT outputs both the estimated diffusion constant and the one obtained from the input B_0 and L_e^{-1} in the terminal. By comparing them, one can evaluate the eventual trade-off between diffusivity and total attenuation.

Chapter 3

Scattering attenuation: *peak delays*

Timeline in papers.		
Paper	**Descriptor**	**Where**
(Saito et al., 2002)	Markov approximation	Link
(Takahashi et al., 2007)	Peak-delay regionalisation	Link
(Takahashi et al., 2009)	Peak-delay tomography	Link
(Calvet et al., 2013)	Peak delay regionalisation + Qc: crust	Link
(De Siena et al., 2016)	Peak delay + Qc: volcano	Link
(Sato, 2016)	Peak delays - short and long λ	Link

3.1 Parameters

Input Parameters for MuRAT		
Symbol	**Descriptor**	**Where**
t_{min}	minimum peak delay allowed	Set in input file
t_{max}	maximum peak delay allowed	Set in input file
t_0 (s)	Origin time	SAC Header

3.2 The peak delay method in pills

The method measures the delay of the maximum energy of the envelope of the direct wave to model scattering attenuation with regionalisation.

Pros of measuring scattering attenuation using peak delays

1. Single-station measurements - you can get all the info from a single seismogram.
2. Proven sensitivity to structural information and geologic boundaries.

Cons of measuring scattering attenuation using peak delays

1. Assumptions underlying the Markov approximation could be unfulfilled. *Tackled by MuRAT estimating parameters at different frequencies and by association with rock-physics research.*
2. No inversion, only regionalisation. *Still not tackled by MuRAT, because often the peak corresponds to trapped S-wave phases.*

Figure 3.1 is a visual representation of the method.

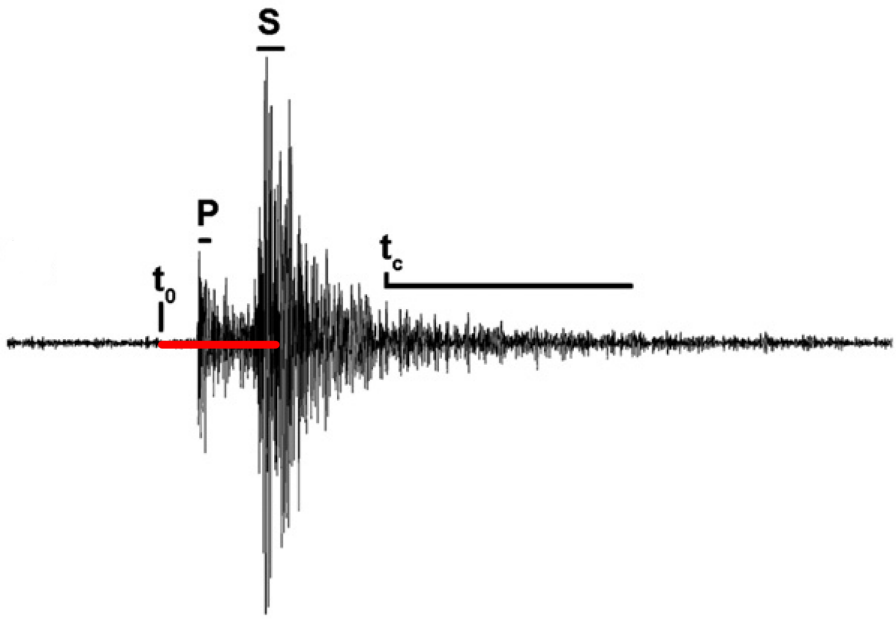


Figure 3.1: The PD method measures the delay of the envelope's maximum from either the arrival time or the origin time of the event.

3.3 The theory bit

While travelling, direct wave packets broaden because of scattering and diffraction. The seminal works for the application of *peak delay* (envelope broadening) imaging are Aki and Chouet (1975), which demonstrates that the coda of high-frequency seismograms originates from random inhomogeneity in elastic properties, and (Sato, 1989), whose first proposed peak-delays as a tool to characterise random heterogeneity. Fehler et al. (2000) proved that the Markov approximation for the parabolic wave equation can synthesise seismic waveforms in two dimensions, including early and intermediate coda. Another seminal work for the technique is Saito et al. (2002), who developed the equations for forward modelling envelopes for spherically outgoing media characterised by von Kármán power spectral density functions (PSDF).

Saito et al. (2002) provides both a historical view and a discussion of the assumptions underlying the method:

- The primary assumption of the method is that the wavelength ($\lambda = 2\pi/k$) is smaller than the

correlation distance (a), so if this is multiplied by the wavenumber: $ak >> 1$. The estimation of a is challenging, but it appears likely that the assumption will be broken at low frequencies, where *diffraction* takes over.

- Propagation is entirely constrained within a single parallelepiped of average velocity V_0. The velocity field depends on space (r) as:

$$V(r) = V0(1 + \xi(r)) \tag{3.1}$$

where $\xi(r) << 1$ are random velocity fluctuations. For a wider description of how to derive auto-correlation and power spectral density functions describing fluctuation distributions, refer to Sato et al. (2012). These quantities are controlled by ϵ (the root-mean-square of velocity fluctuations), a (the correlation length) and κ (tuning the richness of short-wavelength components of the random medium).

- Frequencies are above 1 Hz; below that, the wavelength grows to a level where most assumptions are invalid in lithospheric settings.

The original definition models envelope duration (t_q) as the quantity related to the characteristic time of envelope broadening (t_M) (Saito et al., 2002). It is instructive to look at the definition of t_M given by Sato (2016):

$$t_M(\kappa, \zeta, k_c, r_0) = \frac{\epsilon^2}{2V_0 a} C_L(\kappa, \zeta, k_c) r_0^2; \tag{3.2}$$

where ζ is a number varying between 0.25 and 1.75. Sato (2016) defines it as a tuning parameter and separates random media between long- and short-scale components so that the correlation distance for the small-scale component is: $a_S^{-1} = \zeta k_c$.

This quantity fully defines scattering in the case of long wavelength heterogeneities, depending on average velocity (V_0), hypocentral distance (r_0, which in a homogeneous medium is the product of V_0 and travel time of the phase t), correlation distance (a), and mean squared velocity fluctuations (ϵ^2), the wavenumber, depending on the angular frequency (ω_c) as $kc = \frac{\omega_c}{V_0}$ and where:

$$C_L(\kappa, \zeta, k_c) = \begin{cases} \frac{\pi^{1/2}\Gamma(\kappa+\frac{1}{2})}{(2\kappa-1)\Gamma(\kappa)}(1 - (\zeta a k_c)^{1-2\kappa}) & \kappa \neq \frac{1}{2} \\ 2ln(\zeta a k_c) & \kappa = \frac{1}{2} \end{cases} \tag{3.3}$$

The dependency on V_0 and a seem straightforward until we do not make r_0 and $C_L(\kappa, \zeta, k_c)$ explicit. Let's see what happens for $\kappa = \frac{1}{2}$:

$$t_M(\kappa = \frac{1}{2}, \zeta, k_c, r_0) = \frac{\epsilon^2}{2V_0 a} V_0^2 t^2 2ln(\zeta a k_c) = \frac{\epsilon^2 V_0 t^2 ln(\zeta a k_c)}{a}; \tag{3.4}$$

in which case:

- The characteristic time increases quadratically with travel time, hence the need for correcting this dependency to focus on scattering.
- A sure way to increase scattering is to increase ϵ^2, even if these variations are generally limited.
- Characteristic times increase with frequency and ζ although only logarithmically.
- For actual ranges of correlation distances, t_M can change dramatically.

3.3.1 Applicability

The important assumption for the applicability of peak-delay analysis is that long-scale heterogeneity components produce scattering. To check the validity, one must always test the following conditions: $\zeta a k_c >> 1$. In general, this means a frequency above 1 Hz. In this case, we can rewrite:

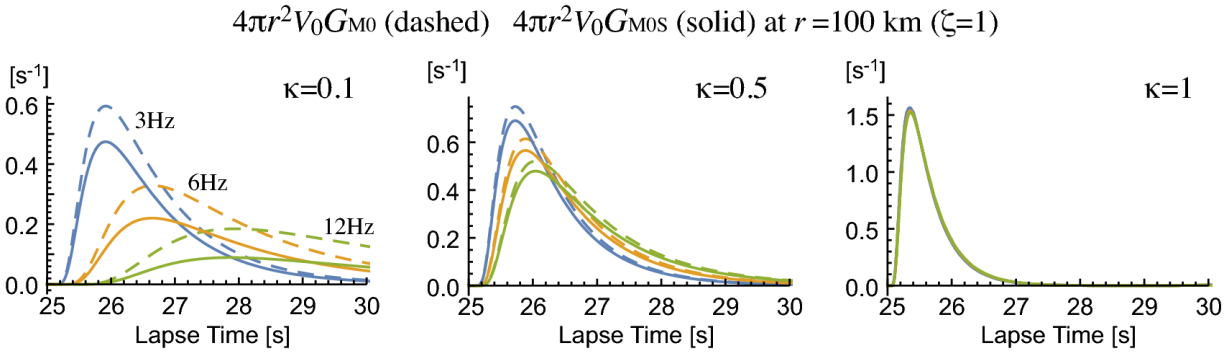


Figure 3.2: Broadening for different κ values. For parameters used, see Sato (2016).

$$t_M(\kappa,\zeta,k_c,r) = \begin{cases} \frac{\epsilon^2}{2V_0 a} r^2 \frac{\pi^{\frac{1}{2}}\Gamma(\kappa+\frac{1}{2})}{(1-2\kappa)\Gamma(\kappa)} (\zeta a k_c)^{1-2\kappa} & \kappa < \frac{1}{2} \\ \frac{\epsilon^2}{2V_0 a} r^2 ln(\zeta a k_c) & \kappa = \frac{1}{2} \\ \frac{\epsilon^2}{2V_0 a} r^2 \frac{\pi^{\frac{1}{2}}\Gamma(\kappa+\frac{1}{2})}{(2\kappa-1)\Gamma(\kappa)} & \kappa > \frac{1}{2} \end{cases} \tag{3.5}$$

This relationship shows the lack of changes with frequency for $\kappa > \frac{1}{2}$ and the slow dependency for $\kappa = \frac{1}{2}$. Characteristic times, envelopes and scattering are strongly frequency-dependent when $\kappa < \frac{1}{2}$. This gives us a quick test of our data. Do waveforms broaden dramatically at high frequencies (Fig. 3.2)? This is already an indication of the richness in long wave numbers (short-scale components). The first check the user should do is then just visual, on the data.

3.4 MuRAT workflow

MuRAT works in analogy to Takahashi et al. (2007), Tripathi et al. (2010) and Calvet et al. (2013) when mapping peak delays. These authors show that above 1 Hz, one must check the relationship between hypocentral distance and measured peak delay. We offer two examples of different hypocentral distances in Figures 3.3 and 3.4. There is a threshold over which you can apply the approximation. Hence Murat shows you the same plot for your dataset - we will discuss in Chapter 9.1.1 the examples at Mount St. Helens, Vrancea and Toba. The peak-delay method requires the same ray tracing approach used for the CN method, discussed computationally in Chapter 2.2.

Following Takahashi et al. (2007), measurements above (below) the logarithmic, linear trend shows high (low) scattering. This version of MuRAT regionalises this procedure. It means positive and negative variations are allocated via their average over blocks after raytracing. The hit count of each segment in each block weights the variation.

3.4.1 Beyond forward scattering

The applicability of peak-delay as imaging attribute has long been hindered by the presence of drastic changes in the scattering regimes of areas of intense heterogeneity, like fracture networks. The basic assumption that long-scale heterogeneity dominates scattering is unfulfilled in these cases, but the log-linear behaviour of peak delays remains a stable features from collisional orogens (Borleanu et al., 2023) down to small-scale volcanic media (Di Martino et al., 2022). Recent research has thus focused

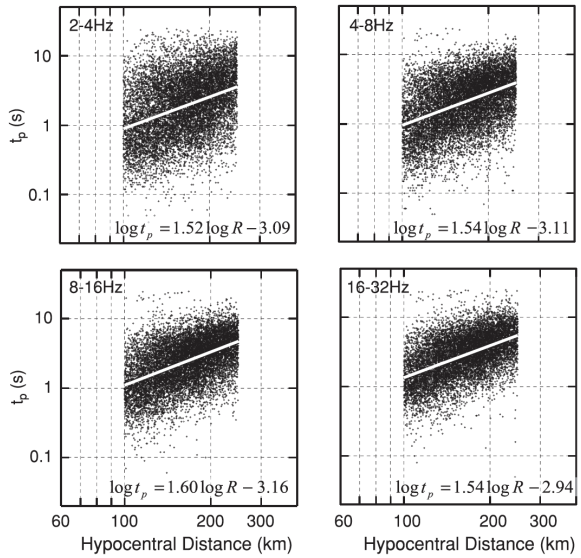


Figure 3.3: Logarithmic plot of peak delay times against hypocentral distances. Black dots represent the data used in this study. White lines are regression lines. From Takahashi et al. (2007).

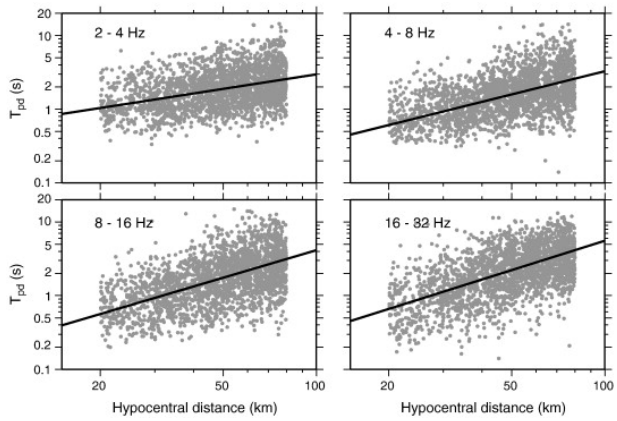


Figure 3.4: Logarithmic plot of peak delay times (in seconds) as a function of the hypocentral distance (in kilometres) for crustal S waves in four frequency bands. Grey dots are the data, and black lines are the regression lines: log10(Tpd) = Ar(f) + Br(f)log10R. From Calvet et al. (2013).

on peak delays at the laboratory scale, demonstrating that fracture network produce trapped waves that drastically increase peak delays at stations nearest to the network (King et al., 2022, 2023). Deformation experiments on sandstone samples show that by increasing strain we increase peak delays

and enable imaging of fractures across a developing shear zone (Fig. 3.5) (King et al., 2022).

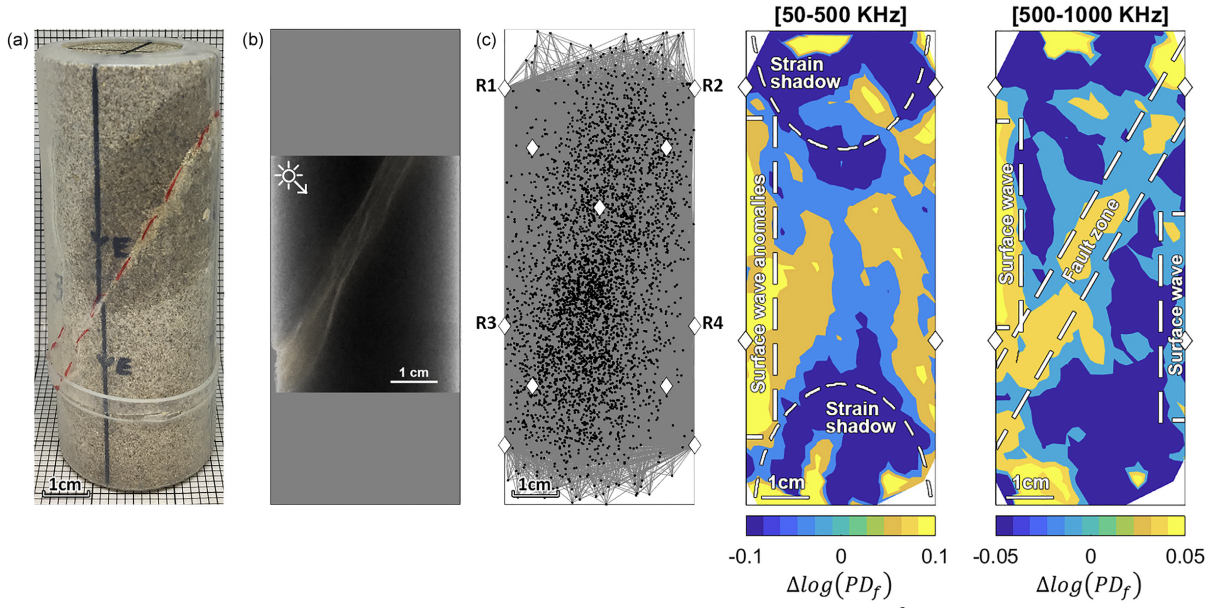


Figure 3.5: Acoustic emission experiment and retrieval of fault zone structure through peak delay imaging, redrawn from King et al. (2022).

The interpretation of peak delay results in MuRAT4 relies heavily on these observations and the corresponding modelling (King et al., 2023). Anomalous high peak delays are due to waves trapped within the rock fractures, developing secondary impulses mixed within the early coda (Fig. 3.6). However, at the field scale, filtering across the right frequency band to image the network is essential. Even at the field scale, fractures developing across networks remain invisible if frequency are too low (wavelengths are too large). In these cases, high-strain, low-peak-delay volumes appear as the primary anomalies (King et al., 2022). If used properly, depending of frequency, researchers can detect areas of high strain accumulation where earthquakes successively nucleate (Gabrielli et al., 2023).

MuRAT4 uses a regionalisation method where peak delay values are weighted by the length of the segment in each cell. In the presence of strong impedance contrasts, sensitivity can diverge dramatically due to trapped waves and resonance in unconsolidated materials (De Siena et al., 2016; Gabrielli et al., 2020; King et al., 2022, 2023). It is thus essential to interpret the maps with complementary tectonic information (Gabrielli et al., 2023; Talone et al., 2023; Napolitano et al., 2025; Talone et al., 2025).

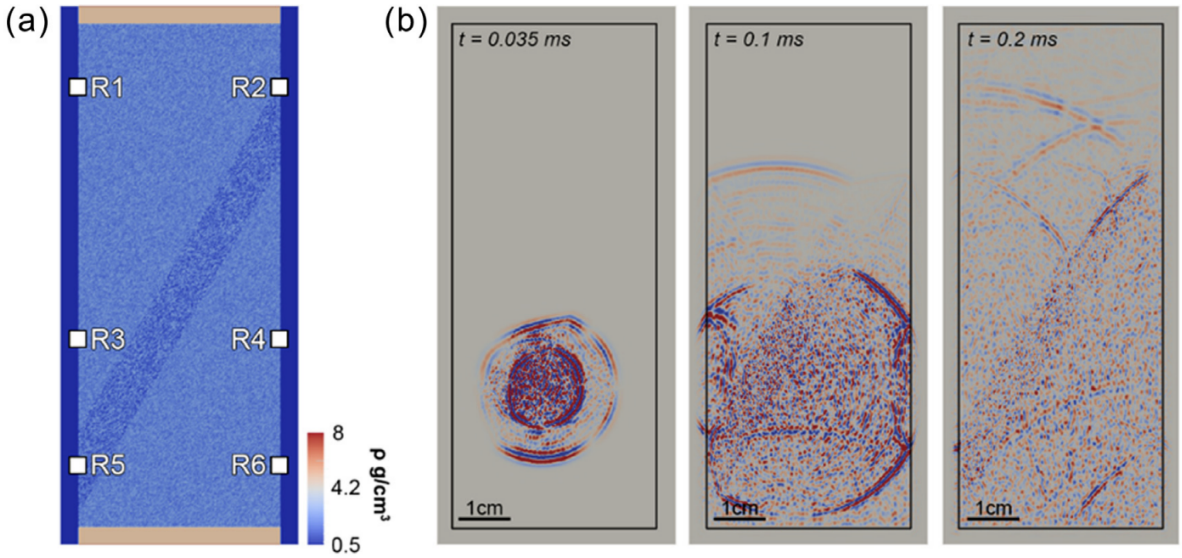


Figure 3.6: Simulation of the wavefield trapped within the fault network structure and increasing drastically peak delays at near-fault receivers, redrawn from King et al. (2023).

Chapter 4

Absorption: *coda attenuation*

Timeline in papers.

Paper	Descriptor	Where
(Aki and Chouet, 1975)	Origin of Coda	Link
(Fehler et al., 1992)	MLTWA	Link
(Calvet et al., 2013)	Peak delay regionalisation + Qc: crust	Link
(De Siena et al., 2016)	Peak delay + Qc: volcano	Link
(Del Pezzo et al., 2016)	2D RTT sensitivity kernels: analytic	Link
(De Siena et al., 2017a)	2D inversion with analytic kernels	Link
(Del Pezzo et al., 2018)	3D RTT sensitivity kernels: diffusive	Link
(Akande et al., 2019)	3D inversion with diffusive kernels	Link
(Sketsiou et al., 2020)	Absorption tomography	Link

4.1 Parameters

Input Parameters for MuRAT		
Symbol	**Descriptor**	**Where**
t_k	starting lapse time	Set in input file
t_w	coda window length	Set in input file
t_0 (s)	Origin time	SAC Header

4.2 The Qc method in pills

The method measures the envelope's decay from a given lapse time to model absorption with an inversion procedure.

Pros of measuring absorption using Qc

1. Single-station measurements - you can get all the info from a single seismogram.
2. Proven sensitivity to fluids and hints at sensitivity to temperature.
3. High-resolution information on near-source and near-station structures.
4. Inversion procedure using differential diffusive sensitivity kernels.

Cons of measuring absorption via Qc

1. The diffusive assumption is rarely fulfilled, with the onset of surface waves and trapped wave especially corrupting low frequencies. *Tackled by MuRAT estimating parameters at different frequencies and testing different coda windows.*

2. Assumption of single highly-diffusive layer rarely fulfilled. *Cross-checking with peak-delay results.*

Figure 4.1 visually represents the method.

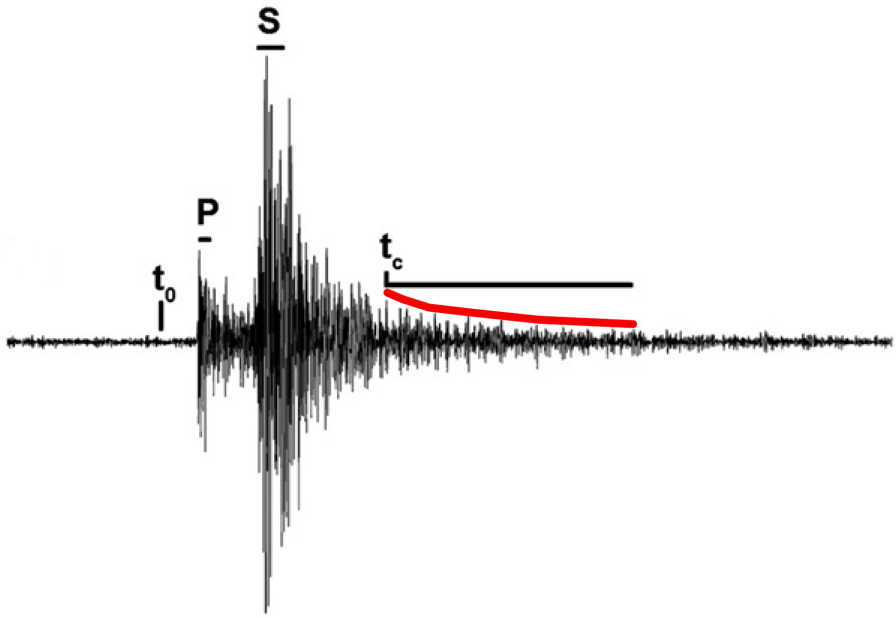


Figure 4.1: The Qc method measures the envelope's decay from a given lapse time from the origin time of the event.

4.3 The theory bit

4.3.1 MLTWA

Coda waves can be used for more than just normalising direct wave energies. Starting from the seminal work of Aki and Chouet (1975), coda waves have been used to characterise the tectonic state of the Earth. This has led to the development of the Multiple Lapse Time Window Analysis (MLTWA) technique Fehler et al. (1992) to separate scattering attenuation Q_s^{-1} from absorption Q_i^{-1}. It does so by measuring extinction length ($L_e = Q_s^{-1} + Q_i^{-1}$) and seismic albedo ($B_0 = \frac{Q_s^{-1}}{L_e}$).

Two assumptions are necessary to apply the MLTWA technique: the direct S-wave dominates the early portion of an S-wave seismogram; the S-coda comprises scattered S-waves. While the total

attenuation of the medium controls the first S-wave amplitude, coda-waves are a product of scattering (Sato et al., 2012). The filtered seismograms are divided into three time-windows, starting from the S arrival, defined by a starting time t_k and window length t_w. The time integrals $E_k(r)$ of the seismic energy density $E(r,t)$ in each window k is:

$$E_k(r) = \int_{t_k}^{t_k+\Delta t} E(r,t)dt \quad k = 1...3 \tag{4.1}$$

where the energy is still a function of the source-station distance *r*. The wave energy decay for each cycle $-\Delta E/E$ is a function of the total quality factor Q (related to the attenuation coefficient η) via the following equation:

$$-\frac{\Delta E}{E} = \frac{2\pi}{Q} = 2\pi\left(\frac{1}{Q_i} + \frac{1}{Q_s}\right) = \frac{\upsilon\eta}{f} \tag{4.2}$$

where υ is the average wave speed and f is the frequency, respectively. The next step consists of correcting the energy of each window $E_k(r)$ for both the geometrical spreading and the integral of the t_w-seconds-long normalisation window (the last window in time). This procedure removes source and site effects (Mayeda et al., 1992):

$$E_k^{obs}(r) = \log_{10}\left(4\pi r^2 \frac{E_k(r)}{\int_{t_{coda}}^{t_{coda}+\Delta t} E_k(r,t)dt}\right) \tag{4.3}$$

The resulting logarithms are plotted versus distance in the four frequency bands considered and fitted to the theoretical normalised energies. The theoretical curves are modelled using the L2-norm misfit function for several waveforms from i to N:

$$M(L_e^{-1}, B_0) = \sum_{i=1}^{N}\sum_{k=1}^{3}[E_k^{obs}(r_i) - E_k^{theo}(r_i, L_e^{-1}, B_0)]^2 \tag{4.4}$$

$E_k^{theo}(r_i, L_e^{-1}, B_0)$ is the theoretical normalised energy computed at distance r_i for the k-th time-window, fixing the L_e^{-1} and B_0 obtained from MLTWA. M is minimised with a grid search in the B_0 and L_e^{-1} parameter space, and the best-fit values $\hat{L}_e^{-1}$ and $\hat{B}_0$ are given by the minima of the function. The isolines of the variable $M_{norm} = M(L_e^{-1}, B_0)/M(\hat{L}_e^{-1}, \hat{B}_0)$ give the error estimates on $\hat{L}_e^{-1}$ and $\hat{B}_0$. Being M_{norm} an F-variable (Del Pezzo and Bianco, 2010), the model parameters can

be estimated using an F distribution (Pisconti et al., 2015), with a level of confidence higher than F=0.68.

4.3.2 Sensitivity kernels

Due to the increased lateral sensitivity compared to ray-dependent techniques, coda waves are theoretically the ideal tool to image the Earth. In practice, apart from the cases of single scattering and diffusion, there is no simple analytical solution, and envelopes have to be dealt with Radiative Transfer Theory (Sato et al., 2012), possibly mixed with ray-tracing techniques that model direct wave information (Sanborn et al., 2017). However, if the wave field is diffusive, one can obtain coda sensitivity kernels with Paasschens equations (Paasschens, 1997) and the approximations of Pacheco and Snieder (2005). Several authors have obtained 2D diffusive kernels based on Radiative Transfer Theory (Obermann et al., 2013; Margerin, 2013; Mayor et al., 2014; Del Pezzo et al., 2016). These kernels have maximum sensitivity at source and receiver (they show a pole at their exact location that must be interpolated) and are all very similar (see, e.g. Figs. 4.2) despite being obtained with different approximations. They primarily aim at coda-wave interferometry but have found applications in coda-attenuation imaging at regional and local scales (Mayor et al., 2016; De Siena et al., 2017a; Napolitano et al., 2020; Sketsiou et al., 2020; Reiss et al., 2022; Napolitano et al., 2025). (Calvet et al., 2013) first noticed the contribution of surface waves to low-frequency observations, an effect modelled in 2D by Obermann et al. (2013) and fit on data by Gabrielli et al. (2020). Del Pezzo et al. (2018) have extended the formulation to 3D media, setting the spatial and frequency limits of the kernel applicability and discussing the assumptions behind the applicability of the kernels.

There are two published approaches to map coda attenuation in the 3D space with kernels. The first is regionalisation (Del Pezzo et al., 2018) in analogy to peak-delay mapping. An excellent review paper on this approach is Del Pezzo and Ibáñez (2020). MuRAT follows instead the pathway laid down by De Siena et al. (2017a). It assumes that all coda energy is contained within the propagation grid at the chosen lapse time. If true, coda attenuation in space can be obtained from source-station measurement with a standard inversion. Sketsiou et al. (2020) extensively discusses the differences between regionalisation and inversion. Akande et al. (2019) produced the first 3D kernel-based coda attenuation model obtained with 3D sensitivity kernels and an inversion approach. This is the approach used by MuRAT3.

As anticipated in Chapter 1, the combined effect of coherent and incoherent scattering as well as the trade-off induced by changing scattering regimes (from Rayleigh to Mie scattering, Cormier and

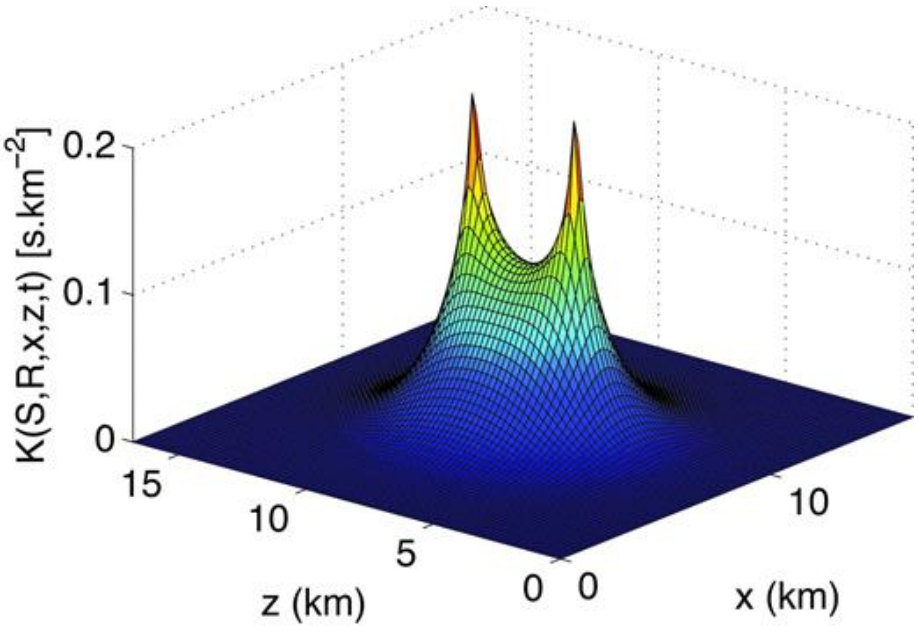


Figure 4.2: Diffusive sensitivity kernels at time 3.6 s obtained by Obermann et al. (2013)

Sanborn (eg 2019)) can profoundly affect the reliability of the kernels. For these to be applied, there is a need for a volume with uniform (or gradually changing) scattering properties, which is rarely valid at all frequencies investigated. Surface Obermann et al. (2013) and resonant Margerin (2013); De Siena et al. (2013) waves, sharp horizontal transitions in scattering (Wegler, 2005; Margerin, 2017; Nardoni et al., 2021) and more generally not accounting correctly for the trade-offs with geometrical spreading (Morozov, 2011) all bring to inexact, if not incorrect, images of the Earth's scattering and absorption properties.

4.3.3 Applicability

MuRAT measures coda attenuation in the diffusive assumption; hence one must check that Q_c^{-1} does not vary with hypocentral distance (Calvet et al., 2013). However, this only tests that equipartition is in place, a necessary but insufficient condition for diffusion. The analysis of coda attenuation with frequency is central to testing the applicability of both MLTWA and sensitivity kernels (Sketsiou et al., 2020). Also, as we are using the inversion procedure, it is paramount to test that most coda energy remains inside the inversion grid (Akande et al., 2019).

MuRAT offers tools to test these approximations and the effect of Qc on measurements of total attenuation (chapter 2).

4.3.4 MuRAT workflow

MuRAT works in analogy to Mayor et al. (2016), De Siena et al. (2017a) and Akande et al. (2019) when modelling coda attenuation. These authors show that above 1 Hz, one can check the relationship between hypocentral distance and measured coda attenuation to test equipartition. In Figure 4.3, we show the expected relation of coda attenuation versus hypocentral distance for the theory's applicability and interpretation of Qc as an absorption measurement. It is clear that the standard assumption of twice the S-wave travel time (Aki and Chouet, 1975) does not work. Hence, MuRAT shows you the same plot for your dataset. However, the ideal threshold (100 s) removes too many data for reliable coverage. Figures 4.3c shows the acceptable compromise.

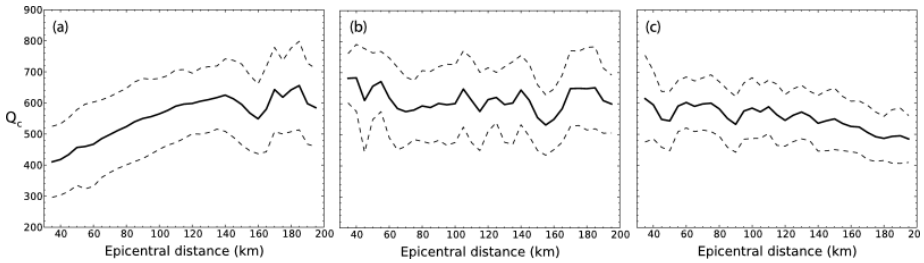


Figure 4.3: (a) Qc as a function of the epicentral distance in the 4–8 Hz frequency band. The duration of the coda window is fixed ($t_w = 50$ s). The solid and dashed lines show the mean dependence of Qc with the epicentral distance and the associated uncertainties, respectively. The analysis is performed on the onset of coda window t_k equal to (a) 2tS, (b) 100 s, and (c) 70 s. From Mayor et al. (2016)

MuRAT also outputs the sensitivity kernels in two grids: (1) a fine propagation grid, extending generally well beyond the area of inversion, and (2) the inversion grid. The first is to compute the kernels and interpolate the poles at source and receiver locations (Akande et al., 2019). The user chooses the boundaries of the second depending on source-station coverage. The code shows what percentage of the energy is still inside the inversion grid.

4.4 MuRAT4 assumptions

MuRAT3 adopted absolute sensitivity kernels to map Qc in space. With increasing lapse time, coda energy is increasingly sensitive to space around source and receiver; however, most of the sensitivity remained at the source and receiver locations (Del Pezzo et al., 2018). Thus normalization across the inversion grid and interpolation at the exact source and receiver location are enough to account for relative energy variation.

MuRAT4 more properly tackles the fact that we measure coda-wave energy across a limited high-lapse-time window (Fig. 4.1). In this case we need to define *differential kernels* between the starting- and end-time of the window, t_s and t_e. The differential diffusive energy between these lapse times is:

$$E_D(f, t_{e,s}) = \frac{S_S(f)G(f)I(f)}{4\pi D}[t_e^{-1.5}\exp\left(-\frac{r^2}{4Dt_e} - 2\pi f Q_I^{-1} t_e\right) - t_s^{-1.5}\exp\left(-\frac{r^2}{4Dt_s} - 2\pi f Q_I^{-1} t_s\right)]. \tag{4.5}$$

The corresponding sensitivity to space is obtained by subtraction of the sensitivity kernel at time t_e and t_s. Fig. 4.4 shows the absolute kernels calculated on a fine grid (left) and the differential (right) sensitivity kernels...

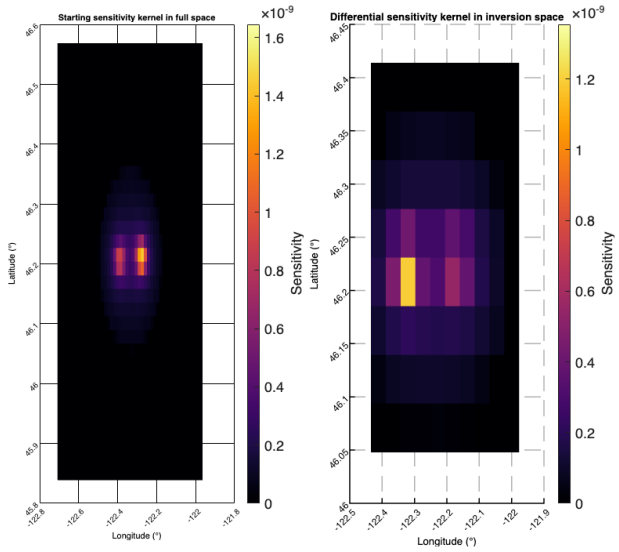


Figure 4.4: (a) Absolute sensitivity kernels measured at the starting time of the coda window $t_s = 15$ s in the Mount St. Helens case (b) Differential sensitivity kernel between $t_e = 25$ s and $t_s = 15$ s.

Chapter 5

Getting started

5.1 Installation and set up in a nutshell

The current version of the code works following these steps:

1. Download or clone the package at the corresponding GitHub page;
2. Work in the downloaded folder after copying it to an appropriate location on your system;
3. Open one of the three input **.m** files, providing a step-by-step explanation of all inputs ($Murat_inputMSH.m$, $Murat_inputRomania.m$, or $Murat_inputToba.m$) and create your own;
4. MuRAT works with SAC files that must be stored in a single folder and corrected for the instrument function. The files must have populated headers, although the code can work using just some header fields (see Chapter 5.2);
5. Run MuRAT.m and select the name of the input file desired.

5.1.1 Requirements and installations in pills:

- *DOWNLOAD:* MuRAT from Github (Fig. 5.1). Ideally, fork or star the code to get updates. This manual explains how to work with GitHub Desktop.
- *SYSTEM:* The program works on Mac, Linux and Windows systems equipped with MATLAB R2021b or higher (back-tested on R2020b and R2021b).
- *NECESSARY TOOLBOXES:* Curve Fitting, Image Processing, Mapping, Signal Processing, Optimization and Global Optimization Toolboxes. The Parallel Computing Toolbox is recommended for speed.
- *EXAMPLE DATASETS:* Three sample datasets (Mount St. Helens, Romania, and Toba) are included in the package, allowing the user to obtain sample models. The datasets work with the input **.m** files that are provided in the working directory.

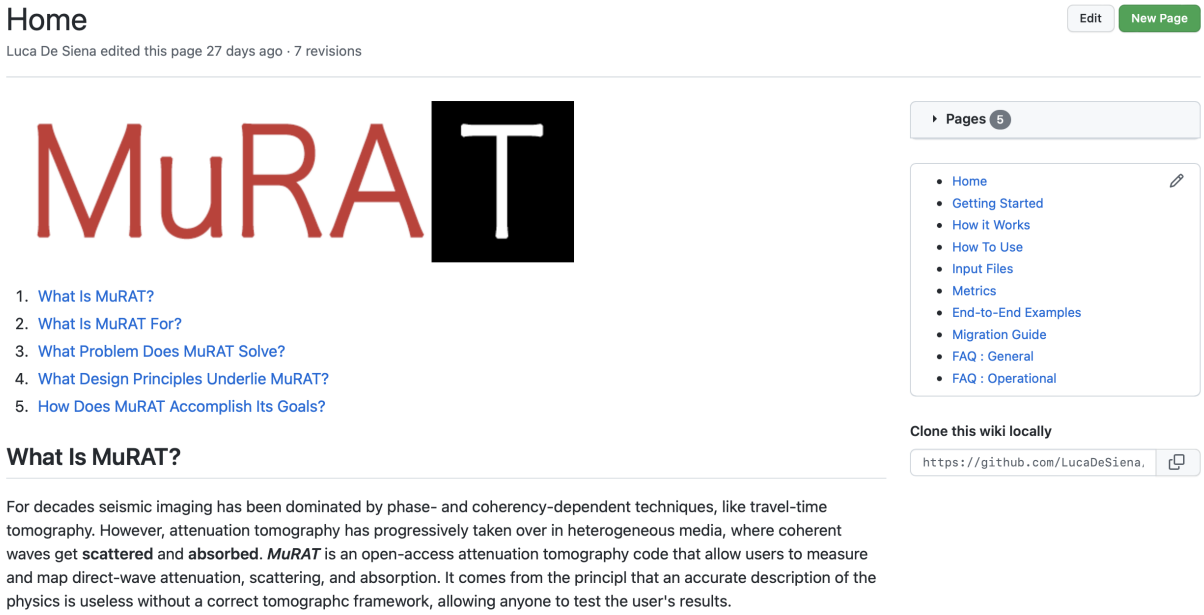


Figure 5.1: The MuRAT page on GitHub.

5.1.2 Example run

We branched the last release of the main code (Fig. 5.1) to our system using GitHub Desktop, then copied the folder to a different location. It is recommended to copy and rename the folder and leave the original MuRAT folder where it is downloaded, in order to highlight and push eventual bugs to main. When finished with your work or if cooperating on that specific dataset with someone, you can then substitute your edited folder in Github. This will allow you to push a new branch that might be used as a repository for codes and results. These repositories are now a requirement for most journals and will enable you to get the seminal version of your results in case of reviews.

The simplest way to start with MuRAT is by reading its readme.txt. This file provides instructions on installing the code and setting up your data, workflow, and outputs. This is the lowest level of documentation but covers the basics without going into jargon or trying to solve all problems at once.

Higher-level documentation is provided by the MuRAT Wiki. Like any wiki on GitHub, it has a lateral panel that allows reaching pages that explain data preparation or input files, for example.

5.2 Data preparation

This section deals with the original SAC files and how to populate their header to work with MuRAT. By populating the header, the user will need no additional file related to data processing.

5.2.1 SAC files

The Seismic Analysis Code has been the standard in seismology for at least 20 years. To process the SAC data in Matlab, MuRAT uses a self-contained MIT-licensed reimplementation of the SAC reader ($fget_sac.m$, MuRAT Authors 2026), whose header struct format is modelled on the original MatSAC tools of Prof. Zhigang Peng (Georgia Tech). The function signature is:

$$[t, data, SAChdr] = fget_sac(pathSACFile) \tag{5.1}$$

where, given the path of the SAC file, the user obtains two variables (*time* and *data*) and especially a structure (*SAChdr*) that stores the fields of the header.

5.2.2 Parameters

The aim of MuRAT has always been to do attenuation imaging using SAC files that can be downloaded from online servers. These data are provided via miniseed that must be unpacked. The resulting SAC might have already populated headers: it is up to the user to populate the parameters shown in the following table if they are blank (or *-12345* as by SAC standard).

Necessary Parameters in SAC header		
SAC field	**Descriptor**	**Name in Matsac header**
o (s)	origin time	$SAChdr.times.o$
a (s)	P-wave pick	$SAChdr.times.a$
t0 (s)	S-wave pick	$SAChdr.times.t0$
stla (deg)	station latitude	$SAChdr.station.stla$
stlo (deg)	station longitude	$SAChdr.station.stlo$
stel (m)	station elevation	$SAChdr.station.stel$
evla (deg)	event latitude	$SAChdr.event.evla$
evlo (deg)	event longitude	$SAChdr.event.evlo$
evdp (km)	event depth	$SAChdr.event.evdp$

5.2.3 Sac Data Folder

The bold-faced parameters in the above table are mandatory. The user can check the parameters in the header of each SAC file using the utilities $Murat_test.m$ and $Murat_testAll.m$ in the **Utilities_Matlab** folder. After running MuRAT the first time, this folder will be added to your path. Otherwise, copy these scripts to your working directory folder.

5.2.4 Murat test for single waveforms

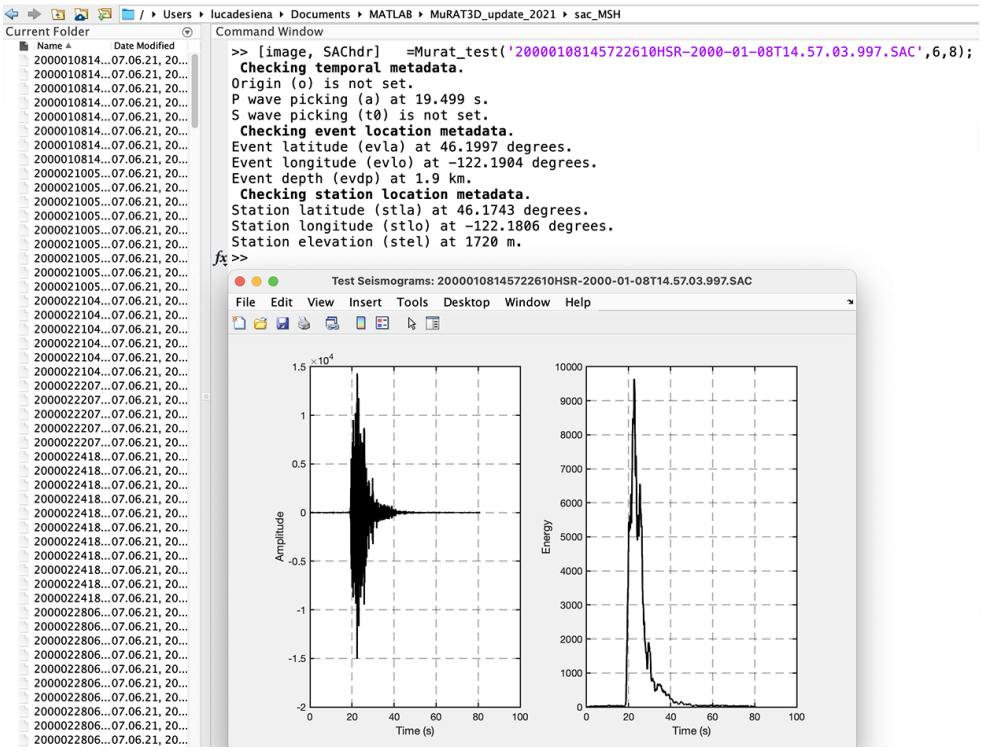


Figure 5.2: Example run of Murat_test. The image is shown while the SACheader is created in the workspace. All the necessary parameters are checked. Notice the missing S-wave picking - the variable storing this parameter may be different from t0.

```
% function [image, SAChdr]    =   Murat_test(nameWaveform,...
%    centralFrequencies,smoothingC,figOutput,verboseOutput)
% TEST seismogram envelopes for changes in broadening
```

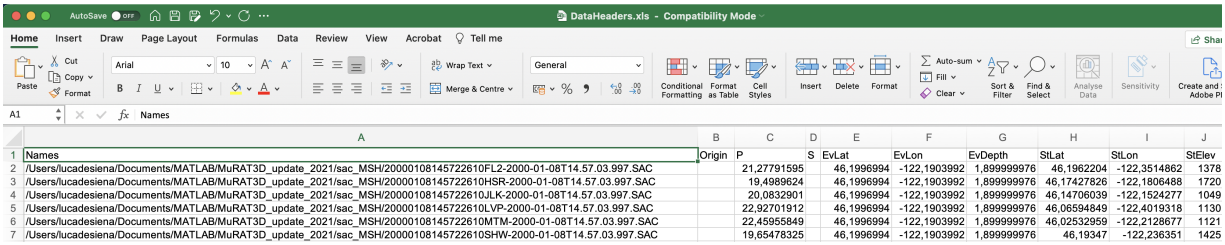
```
% CREATES a figure with seismograms and envelopes for different frequencies
%
% Input Parameters:
%       nameWaveform:           name of the SAC file
%       centralFrequencies:      vector of frequencies (Hz), if [] no filter
%       smoothingCoefficient:   coefficient to smooth envelopes
%       figOutput:                      decide if you want to show figures (set to 1)
%       verboseOutput:             decide if you want to display messages (set to 1)
%
%   Output:
%       image:                          image with the envelope at a specified frequency
%       SAChdr:                         header of the SAC file
%
```

The function uses the path of a waveform, a vector of central frequencies, and a smoothing coefficient as inputs. Two additional flags allow us to show the corresponding seismogram and envelope (set $figOutput = 1$) and the output messages (set $verboseOutput = 1$). It provides the user with the header structure (SAChdr). The fields necessary to apply MuRAT can be checked in the workspace and the command window or displayed directly with the *verboseOutput* on. Figure 5.2 shows a run of Murat_test for a file stored in the sac_MSH folder, filtered at 6 Hz, smoothing eight samples per cycle.

5.2.5 Murat test All, creating a file to check all headers in the dataset

MuRAT checks that all the parameters in the headers for a dataset are populated. It produces error messages for the missing parameters, pointing at the corresponding waveform. Finally, it outputs the Excel file $DataHeaders.xlsx$ in the results, providing an overview of all the required parameters. Fig. 5.3 shows this file after a run over the data folder sac_MSH. Missing parameters will be blank in the file. The origin and S-wave peaking (searched for the SAC variables o and $t0$) are missing. These are optional parameters if using P waves so that the code will work anyway.

The function can be used before running MuRAT. It provides a $DataHeaders.xlsx$ file in the working folder that the user can check to set the variables in the header.



	Names	Origin	P	S	EvLat	EvLon	EvDepth	StLat	StLon	StElev
2	/Users/lucadesiena/Documents/MATLAB/MuRAT3D_update_2021/sac_MSH/20000108145722610FL2-2000-01-08T14.57.03.997.SAC		21,27791595		46,1996994	-122,1903992	1,899999976	46,1962204	-122,3514862	1378
3	/Users/lucadesiena/Documents/MATLAB/MuRAT3D_update_2021/sac_MSH/20000108145722610HSR-2000-01-08T14.57.03.997.SAC		19,4989624		46,1996994	-122,1903992	1,899999976	46,17427826	-122,1806488	1720
4	/Users/lucadesiena/Documents/MATLAB/MuRAT3D_update_2021/sac_MSH/20000108145722610JLK-2000-01-08T14.57.03.997.SAC		20,0832901		46,1996994	-122,1903992	1,899999976	46,14706039	-122,1524277	1049
5	/Users/lucadesiena/Documents/MATLAB/MuRAT3D_update_2021/sac_MSH/20000108145722610LVP-2000-01-08T14.57.03.997.SAC		22,92701912		46,1996994	-122,1903992	1,899999976	46,06594849	-122,4019318	1130
6	/Users/lucadesiena/Documents/MATLAB/MuRAT3D_update_2021/sac_MSH/20000108145722610MTM-2000-01-08T14.57.03.997.SAC		22,45955849		46,1996994	-122,1903992	1,899999976	46,02532959	-122,2128677	1121
7	/Users/lucadesiena/Documents/MATLAB/MuRAT3D_update_2021/sac_MSH/20000108145722610SHW-2000-01-08T14.57.03.997.SAC		19,65478325		46,1996994	-122,1903992	1,899999976	46,19347	-122,236351	1425

Figure 5.3: Example file after running MuRAT for the MSH example. All the necessary parameters are checked. Notice the missing S-wave picking - the variable storing this parameter may be different from t0.

5.2.6 Murat change header - changing headers of single files

The previous analysis could discover some waveforms with missing or wrong header fields. The best procedure is to change these headers direct in SAC; however, users might not be familiar with the code, so $Murat_changeHdr.m$ offers an alternative to change header fields:

```
% function seism                 =   Murat_changeHdr(newfolder)
% CHANGES header of a file
%
% Input Parameters:
%       newFolder:              folder where you save the changed file
%
%   Output:
%       seism:                  new seismogram
%
```

The only input is the path to a folder where the user will store the SAC with a changed header. The user selects the SAC file to be changed through a GUI. Then, in the command window, the user is asked if the event, station or time fields are to be changed. Most often, the problem is with the time. The code assigns the origin to field $'o'$, the P-time to $'a'$, and the S-time to $'t0'$.

Chapter 6

Input files

MuRAT4 offers three input files as examples (Fig. 6.1). A new user needs to edit these files to run the code. The files are self-explanatory, with explanation and figures describing each input. The new user can use this $Murat_inputMSH.m$ file to learn what each parameter does. The input files are divided into sections, whose inputs are described in the following. For a more detailed description, consult the Mount St. Helen input file, shown in Appendix A.

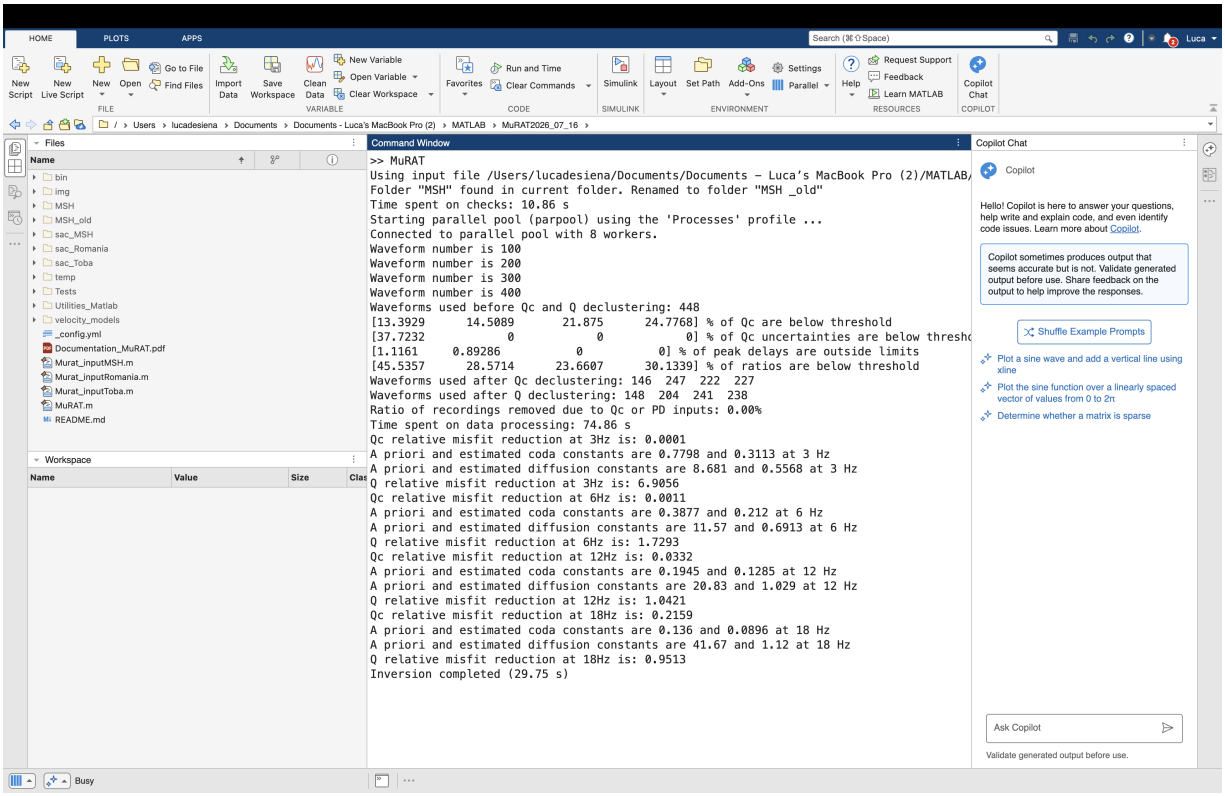


Figure 6.1: Input files are provided as $Murat_input..m$ files. The standard release has three sample inputs (Mount St. Helens - MSH, Romania and Toba caldera).

6.1 The fields of the input

- GENERAL FIELDS:

  $Murat.input.dataDirectory$: Define the name of folders where data are located

  $Murat.input.label$: Defines the name of the folder where results will be stored

  $Murat.input.workers$: Number of cores if parallelised, leave empty for sequential code

- WAVEFORM DATA

  $Murat.input.originTime$: Defines the variable that stores the origin time or leave empty

  $Murat.input.PTime$: Defines the variable that stores the P-time

  $Murat.input.STime$: Defines the variable that stores the S-time or leave empty

  $Murat.input.PorS$: Defines if the analysis is for P- or S-waves

  $Murat.input.centralFrequency$: Defines the set of central frequencies analysed

  $Murat.input.envelopeSmoothTime$: Envelope smoothing window in seconds (1 s if empty)

  $Murat.input.components$: Defines the number of recording components

  $Murat.input.mintravel$: Set the min travel times before which you exclude traces

  $Murat.input.maxtravel$: Set the max travel times after which you exclude traces

  $Murat.input.declustering$: Number used to divide the grid and decluster the dataset

- PEAK DELAY

  $Murat.input.minimumPeakDelay$: Minimum peak delay allowed

  $Murat.input.maximumPeakDelay$: Maximum peak delay allowed

  $Murat.input.stDevPD$: Standard deviation around the increasing linear fit in the test

- DIRECT WAVE ATTENUATION

  $Murat.input.bodyWindow$: Window to measure body waves and noise in seconds

  $Murat.input.startNoise$: Seconds to start noise window from the start of recording

  $Murat.input.tresholdNoise$: Minimum signal-to-noise ratios allowed

- CODA ATTENUATION

  $Murat.input.lapseTimeMethod$: Defines the method used to measure coda waves

  $Murat.input.startLapseTime$: Defines start of coda window

  $Murat.input.codaWindow$: Defines the length of coda window

  $Murat.input.albedo$: Measured Albedo from MLTWA

  $Murat.input.iExtinctionLength$: Measured inverse Extinction Length from MLTWA

  $Murat.input.diffusionConstant$: Decide if the diffusion constant is 'Inverted' or fixed to its 'Theoretical' value

  $Murat.input.kernelTreshold$: Coefficient to make kernels more accurate or quicker to compute

  $Murat.input.QcMeasurement$: Decide if using a 'Linearized' or 'Non-Linear' approach

  $Murat.input.fitTresholdLinear$: Minimum fit accepted in the Linearized case

- GEOMETRY AND VELOCITY

  $Murat.input.origin$: Defines the origin of the spatial grid

  $Murat.input.end$: Defines the end of the spatial grid

  $Murat.input.gridLat$: Number of nodes across latitude

  $Murat.input.gridLon$: Number of nodes across longitude

  $Murat.input.gridZ$: Number of nodes across depth

  $Murat.input.sections$: Location of sections where you will get the maps

  $Murat.input.availableVelocity$: Decides if a 1D (0) or 3D (1) model is available

  $Murat.input.namev$: Defines the name of the velocity model

  $Murat.input.averageVelocityP$: Defines the average P-wave velocity

  $Murat.input.averageVelocityS$: Defines the average S-wave velocity

- INVERSION

  $Murat.input.inversionMethod$: Inverts with Tikhonov regularisation (**'Tikhonov'**) or with the **'Particle'**, **'Annealing'** or **'Global'** (genetic algorithm) optimisers

  $Murat.input.MaximumIterations$: Sets iterations after which iterative optimization will stop

$Murat.input.MaximumStallIterations$: Sets iterations after which iterative optimization stalling will stop

$Murat.input.PlotInversion$: Set to output L curves figures or cost functions

$Murat.input.dampingValueQc$: relative damping for Qc inversion, one value per frequency (0.1 if empty)

$Murat.input.smoothingValueQc$: smoothing for Qc inversion (empty or 0 for none)

$Murat.input.dampingValueQ$: relative damping for Q inversion, one value per frequency (0.1 if empty)

$Murat.input.smoothingValueQ$: smoothing for Q inversion (empty or 0 for none)

$Murat.input.dampingSweepValues$: list of relative damping values for separate inversion runs, or empty for a single run

$Murat.input.sizeCheck$: Decides if doubling or making node space four times in checkerboard

$Murat.input.spikeLocationOrigin$: Origin of the spike block

$Murat.input.spikeLocationEnd$: End of the spike block

$Murat.input.spikeValue$: Value of attenuation/absorption in the block

- PLOTTING

  $Murat.input.PlotRays$: Decides if rays figures are created (1) or not (0)

  $Murat.input.PlotTests$: Decides if test figures are created (1) or not (0)

  $Murat.input.PlotResults$: Decides if result figures are created (1) or not (0)

  $Murat.input.PlotCheckerboard$: Decides if checkerboard test figures are created (1) or not (0)

  $Murat.input.PlotSpike$: Decides if spike test figures are created (1) or not (0)

Each input is necessary, but the user will change only some of them repeatedly. Here, I give details on each group.

6.1.1 GENERAL FIELDS

These are set initially and never changed, except $Murat.input.workers$. The first run should always be sequential (set it to **[]**). The sequential run must be used to debug the code. While MuRAT has several embedded error messages, stopping the run is often necessary, especially during data processing (see $Murat_data$). The debug function in Matlab does not allow pinpointing the incorrect waveform with parallelised processing. However, it is recommended to perform the final runs in parallelised mode: the parallelised processing reduces computational time from ~ 15 s to ~ 4 s for the MSH dataset.

6.1.2 WAVEFORM DATA

You can check the SAC header requirements in chapter 5.2. Once set, the header variable must be the same for all files (e.g., variable *'a'* for P-wave picking or *'o'* for the origin). The number of components is defined at the start. The code allows a single analysis for multiple frequencies with bandwidth $\frac{1}{3}$ of the central frequency. The user can also smooth envelopes with values set in seconds. The user can also decide to exclude all waveforms outside of a certain travel time (e.g., if large travel times correspond to propagation in the mantle, where there is a different scattering regime). Finally, the declustering variable divides the grid into smaller cells to select the best earthquake waveforms in the cell, avoiding different repeated measurements that increase residuals. It is recommended to test the code first without declustering.

6.1.3 PEAK DELAY

See chapter 3 for additional details. Setting the minimum and, especially, the maximum allows avoiding tracking phases unrelated to the direct wave arrival. Sometimes tracking these phases, typically surface waves generated by a strong impedance contrast at the surface, is helpful to map changes in topography and geomorphology (De Siena et al., 2016; Gabrielli et al., 2020). The standard deviation setting helps exclude outliers or secondary trends that appear in the test.

6.1.4 DIRECT WAVE ATTENUATION

See chapter 2 for additional details. $Murat.input.bodyWindow$ must be small enough to assume still a ray approximation but large enough to mitigate changes in radiation pattern, the primary trade-off of the coda-normalisation method. $Murat.input.startNoise$ depends on how much noise you have at

the start of your recordings. The code will measure noise from this point on the window set for the body wave and use it to compute signal-to-noise ratios. All waveforms below $Murat.input.tresholdNoise$ are discarded.

6.1.5 CODA ATTENUATION

See chapter 4 for additional details. There are three methods to measure coda: using a constant start time (**'Constant'**), the peak amplitude (**'Peak'**) and a number multiplied by the travel time (**'Travel'**). In all cases, the start of the coda window is a lapse time measured from the origin time of the event. The standard practice uses the last option (e.g., $2t_S$). However, coherent waves are often still in the corresponding window (see chapter 8) (Calvet et al., 2013). The **'Peak'** is used when employing active data in volcanic, as the waveform becomes rapidly diffusive (Wegler, 2003).

If the user did not measure albedo or extinction length, they could use published values for the area or the crust. $Murat.input.diffusionConstant$ sets how the diffusion constant enters the coda-normalisation method (see chapter 2): **'Theoretical'** fixes it from albedo and inverse extinction length, while **'Inverted'** estimates it from the data and compares it with the theoretical value. $Murat.input.kernelTreshold$ must be set to find a compromise between the accuracy of the kernels and computational costs. Setting it to 1 solves the forward kernels on the same grid used in the inversion: this is typically too coarse, e.g., to interpolate the poles of the kernels at the source and station. The minimum value is set to two, with 4/8 providing the most accurate results.

While the standard practice is to linearise the inversion by taking the logarithm of the envelope, the user can choose a non-linear approach (Napolitano et al., 2020). In the first case, the minimum accuracy is set with $Murat.input.fitTresholdLinear$, representing the Pearson coefficient relative to a line. The advice is to set it to a very low value (e.g., 0.1) as this measurement will be used to weigh the inversion. In the non-linear case, the weight is given by the PDF of the misfit.

6.1.6 GEOMETRY AND VELOCITY

The inversion grid's extension and number of nodes are often changed throughout the analysis because of the need for testing and avoiding spurious earthquakes. MuRAT inverts on a regular grid defined in latitude, longitude and depth. If using a 1D model ($Murat.input.availableVelocity$=0), the user can input depths in km; otherwise, using a 3D model ($Murat.input.availableVelocity$=1), depths are in meters. The text file of the 3D velocity model must replicate the structure of $modvMSH.txt$

for the 1D that of $iasp91.txt$. Both files are inside the **velocity_models** folder. The user can set average velocities for P and S waves for standard calculations.

6.1.7 INVERSION

In MuRAT4, the regtools are attached but now used only to compute L curves when using the Tikhonov method (set as $Murat.input.inversionMethod$ = **'Tikhonov'**). We switched to using both the Optimization and Global Optimization toolboxes, embedding three of the global optimization strategies (the particle swarm (**'Particle'**), simulated annealing (**'Annealing'**), or genetic algorithm (**'Global'**) solvers), included in a MATLAB license. The user dampens the inversion by selecting the corresponding parameter for each frequency; damping is relative to the inversion matrix, and smoothing is available for all methods (see Chapter 10). The other inputs serve to set if intermediate figures showing iterative inversion process appear (i.e. cost function, read the toolbox documentation). A list of damping values in $Murat.input.dampingSweepValues$ runs a separate inversion for each value. In the synthetic testing, the user defines the multiplication factor of the checkerboard test relative to node spacing (2 or 4). The inputs for the spike test define its location and value.

6.1.8 PLOTTING

This section lets the user define which figure must be output. Plotting is very time consuming, this option becomes very useful if, for example, the user only wants to output the measurement tests without the results of the inversion.

Chapter 7

Output files

MuRAT4 has several output folders containing files that give the user insights into the quality of the analysis (Fig. 7.1). Also, after running, the code creates fields related to the output in the $Murat.mat$ file.

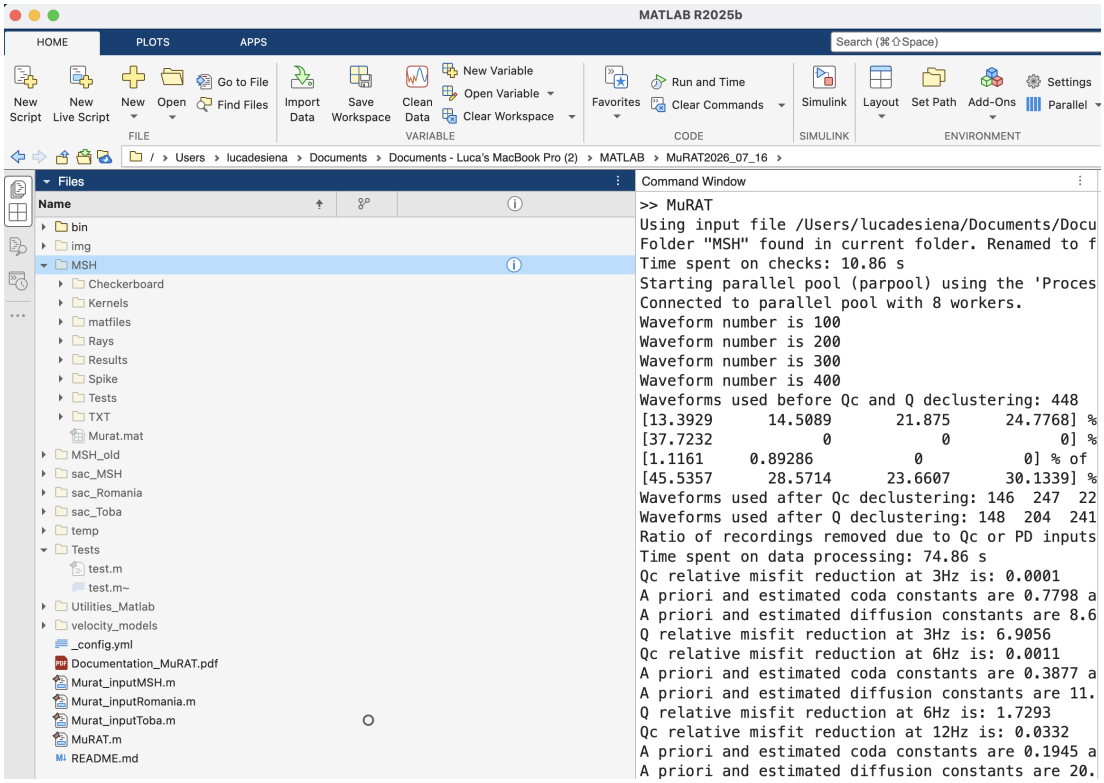


Figure 7.1: Example structure of the results after a run on the MSH dataset.

7.1 The folders and files after running the code

- CHECKERBOARD:

  $PDfolder$: Contains hit maps for the selected sections.

  $Qfolder$: Contains the frequency-dependent .fig and .tif files showing the checkerboard test inputs and outputs for the Q inversion across the sections defined in the input.

  $Qcfolder$: Contains the frequency-dependent .fig and .tif files showing the checkerboard test inputs and outputs for the Qc inversion across the sections defined in the input.

- KERNELS

  The folder contains all the frequency-dependent kernel and differential kernel figures for all methods.

- RAYS

  The folder contains all the frequency-dependent ray figures for all methods.

- RESULTS

  $PeakDelay$: Contains the frequency-dependent .fig and .tif files showing the results of the peak delay analysis across the sections defined in the input. The same figures are provided, removing cells with less than 5% of the total rays.

  Q: Contains the frequency-dependent .fig and .tif files showing the results of the Q analysis across the sections defined in the input.

  Qc: Contains the frequency-dependent .fig and .tif files showing the results of the Qc analysis across the sections defined in the input.

  $Qc_vs_frequency.png$: Figure showing the Qc dependence on frequency

- SPIKE

  $Qfolder$: Contains the frequency-dependent .fig and .tif files showing the spike test inputs and outputs for the Q inversion across the sections defined in the input.

  $Qcfolder$: Contains the frequency-dependent .fig and .tif files showing the spike test inputs and outputs for the Qc inversion across the sections defined in the input.

- TESTS

  $Clustering$: A figure showing the original rays (black) and those selected after declustering (red).

  $LCurve$: A folder that contains the frequency-dependent .fig and .tif files showing the L curves and cost functions for the chosen inversion.

  $PeakDelay$: Test for the reliability of the peak delay method when applied to data.

  Q: Test for the reliability of the Q method when applied to data and Picard condition for the inversion.

  Qc: Test for the reliability of the Qc method when applied to data and Picard condition for the inversion.

  $DataHeaders.xlsx$: File checking the existence of required fields in SAC headers.

  $VTest.fig$: Figure checking the transformation of the 3D velocity model in the chosen reference system.

- TXT

  $peakdelay$: Frequency-dependent text files to show peak delay results. The fourth column is the peak delay, while the fifth is ray crossing.

  Q: Frequency-dependent text files to show Q results. The fourth column is the inverse Q. The fifth is all zeros. The sixth and seventh (eighth and ninth) columns are the inputs and outputs of the checkerboard (spike) tests.

  Qc: Frequency-dependent text files to show Qc results. The fifth is all zeros. The sixth and seventh (eighth and ninth) columns are the inputs and outputs of the checkerboard (spike) tests.

- Murat.mat

  File containing all inputs and outputs of the run.

- DAMPING SWEEPS

  With $Murat.input.dampingSweepValues$ set, each damping value gets its own results folder named after the label and the value (e.g., $Toba_damp_0p1$), containing the same sub-folders and a $Murat_damp_0p1.mat$ file. Kernels, rays and data tests are computed once and copied to each folder. Existing folders with the same names are archived as $_old$, $_old_2$, etc.

7.2 The Murat .mat file

The Murat.mat file saves all information in the $input$, PD, Q, Qc and $rays$ fields. The $input$ field recaps all the input the user has set for the run. The others contain all results obtained during the run, separated by method, plus ray outputs. Among these, the users are likely to check $Murat.problemPD, .problemQ, .problemQC$. These structures include the paths to files that did not satisfy the requirements for the analyses, where each column corresponds to a figure. The user can check or remove these files from the study.

Chapter 8

Tests

The code provides test figures for the three methods. These are all in the **Tests** folder, created after running the code once. These tests are divided into three categories:

- **Analyses** of method data and inversion quality;
- **Curves** that allow defining the best damping parameters for the inversion;
- **Plots** of the behaviour of coda attenuation vs frequency and velocity models.

8.1 Analyses

The *Analyses* figures (files $*.analysis_f_Hz$) are a crucial output in MuRAT. They allow for assessing the data quality and inversion (except for the peak delay, which is only regionalised).

8.1.1 Peak delays

As described in Chapter 3, the hypothesis behind peak-delay imaging is that peaks increase vs travel times. Figure 8.2 shows the $PD_analysis_frequency$ figures for the three sample datasets.

These figures allow you to compare the differences (and problems) from peak-delay analysis in different frequency bands. The Mount St Helens example (Figure 8.1) shows how the 3 Hz frequency is affected by surface waves acting at short travel times. This effect is due to unconsolidated materials from the 1980 debris avalanche (Gabrielli et al., 2020). Analyses are only possible above 6 Hz.

The user needs to check the fit of the peak-delay behaviour to a line to apply the regionalisation procedure, as explained in Chapter 3.

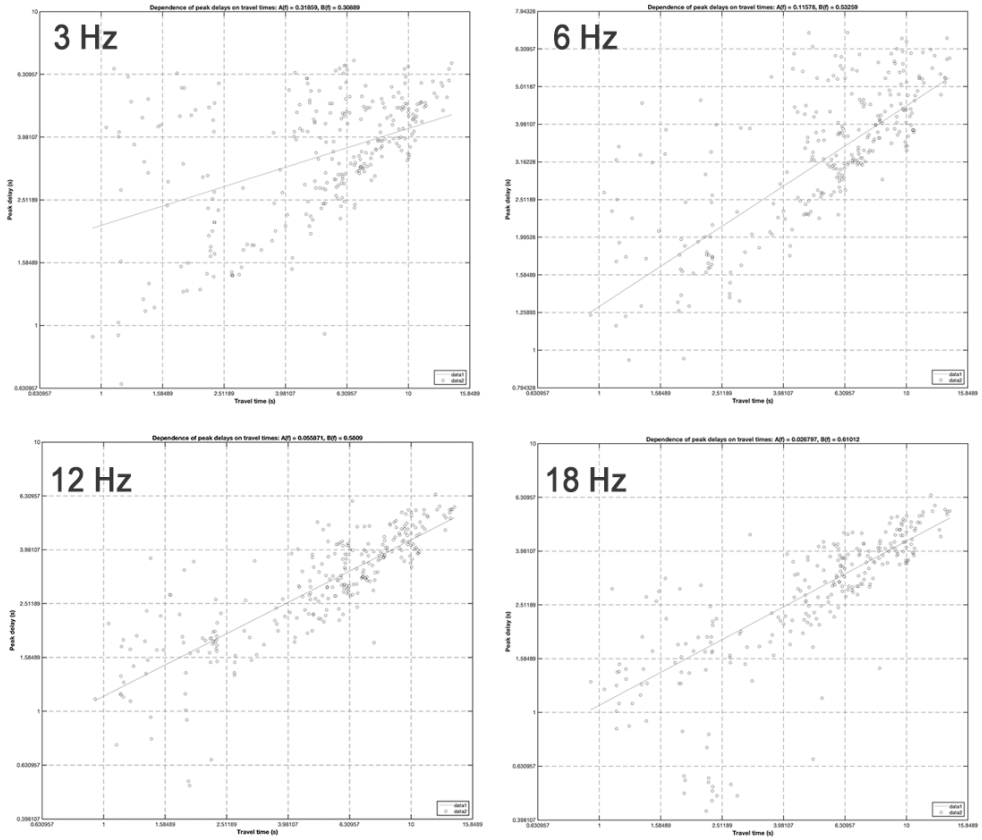


Figure 8.1: Comparison between peak delays for the four frequencies and the Mount St Helens dataset.

8.1.2 Coda attenuation

As explained in Chapter 4, **coda attenuation must be constant with increasing ray length** (Calvet et al., 2013) to interpret values of Q_c as absorption. MuRAT outputs this simple variation for a first-order check on the data. A constant Q_c does not mean the field is diffusive, it only indicates equipartition is. One can use the results of an *MLTWA* analysis (Fehler et al., 1992; Del Pezzo et al., 2001; Del Pezzo, 2008) for better calculations. Codes like Qopen (Eulenfeld and Wegler, 2016) precisely measure mean free path and transport mean free path, providing better constraints on the onset of diffusion. However, their results are generally equivalent to those of MLTWA (van Laaten et al., 2021).

We obtain the Picard condition using the external package regtools (Hansen, 2007). The Picard

condition gives a first glance at how many coda-attenuation parameters will be solved by the inversion in space (see Aster et al. (2005) for a review).

8.1.3 Direct-wave attenuation

The coda-normalisation method (chapter 2) **requires the use of direct waves that decay linearly with travel time (hypocentral distance)**. The coda-normalisation analysis (Fig. 8.4) must show that its normalised logarithm decays for increasing travel time (upper panel), improving the linear decrease behaviour of the uncorrected direct-wave energy The data (black circles) are compared to the analytical forward with average attenuation measured on the data (red stars). The middle panel shows the behaviour of direct energy without normalisation and correction for the velocity model. The inversion is tested with the usual Picard condition, showing that, for the chosen parametrisation, more than half of the parameters are solved.

MuRAT4 now relies on a diffusive approximation for the coda - it actually models the coda behaviour as diffusive. This means that the selected albedo and extinction length play a critical role in the fitting, as they define the underlying diffusion constant. More importantly, the code now highlights when this assumption is unfulfilled. At 3 Hz (Fig. 8.4), the normalization at 3 Hz (upper panel) shows a positive behaviour, opposite to the one we expect from a wave losing energy during propagation, and thus a negative quality factor. To the contrary, the normalization at 18 Hz (Fig. 8.4) (upper panel) produces the expected negative linear behaviour. The anomalous effect at 3 Hz is caused by anomalous data recorded across the debris avalanche, caused by the collapse of the northern part of the crater (De Siena et al., 2016).

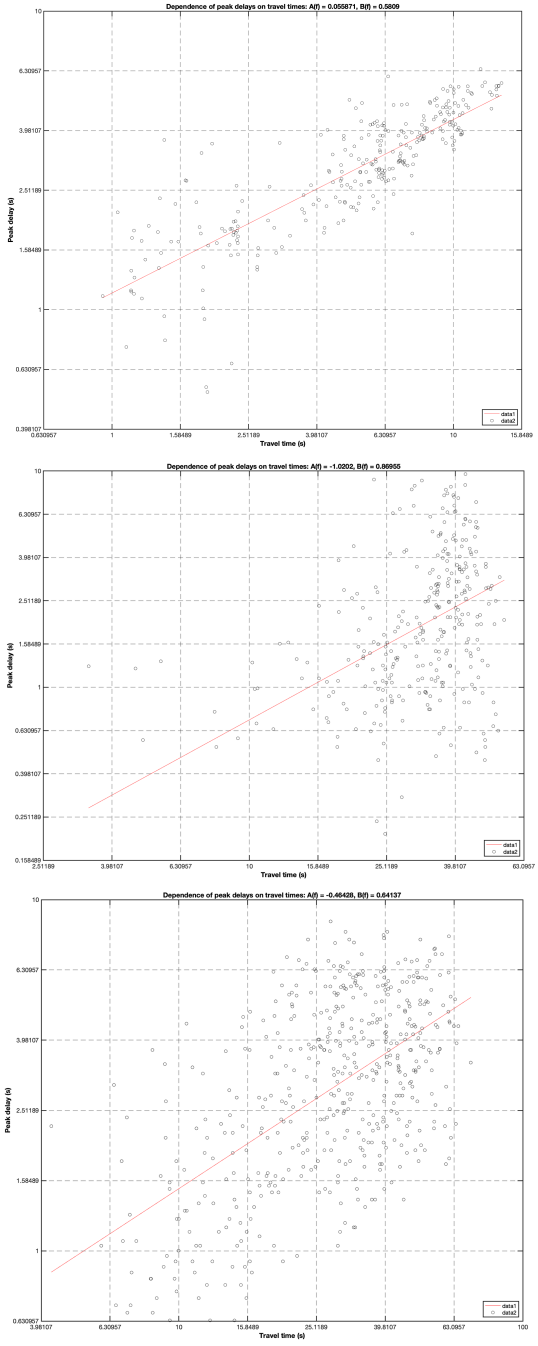


Figure 8.2: Comparison between peak delays for the three sample datasets.

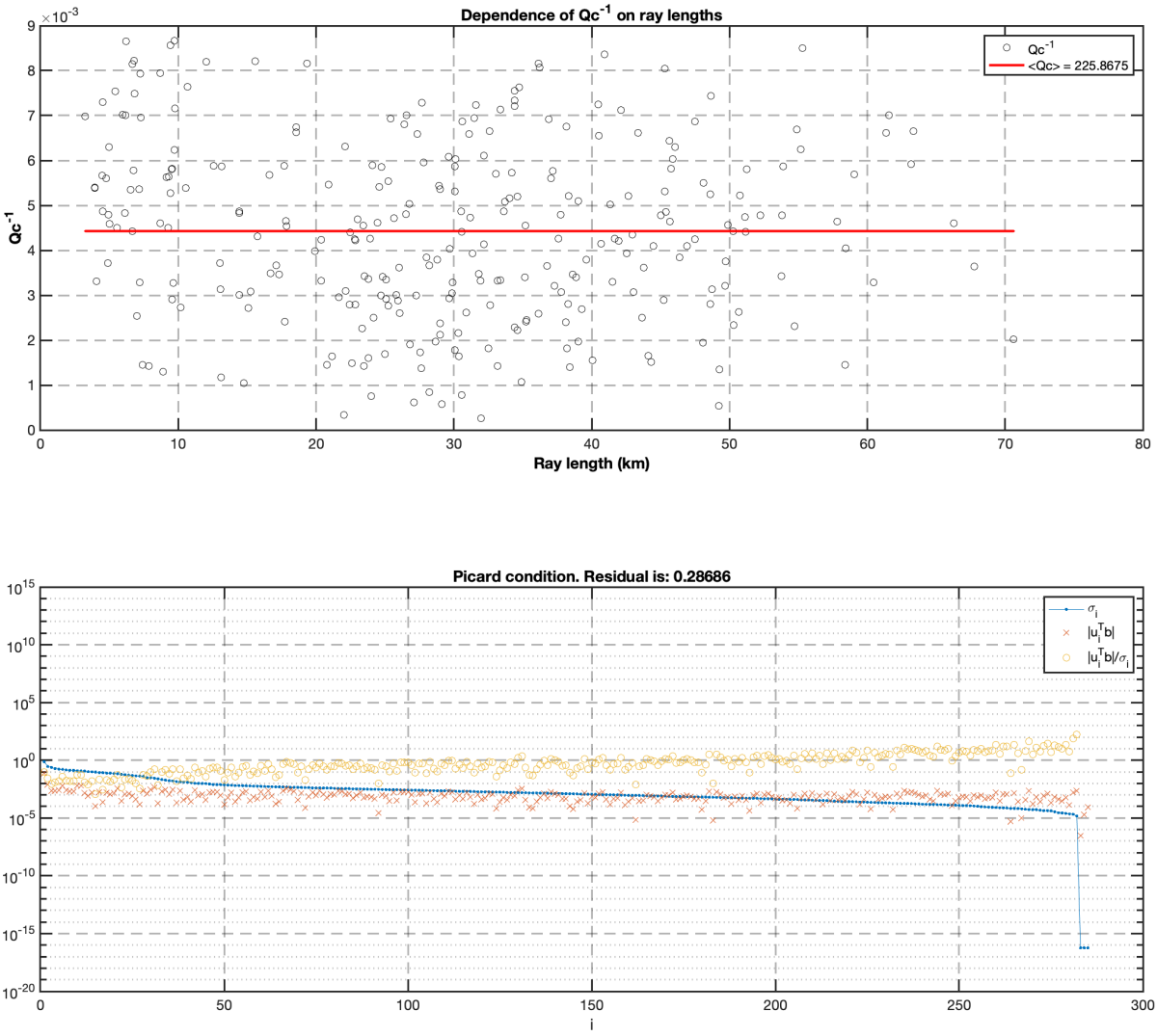


Figure 8.3: Coda analysis plot for the MSH dataset at 6 Hz. The upper panel shows that Q_c is constant with the ray length computed through ray tracing. The lower panel depicts the result of the Picard condition (Aster et al., 2005): only half of the parameter models are solved under the weakest hypothesis (blue and red lines cross around the 150th parameter). Only one-tenth of the parameter models are solved if we also consider the uncertainty of data. σ are the singular values.

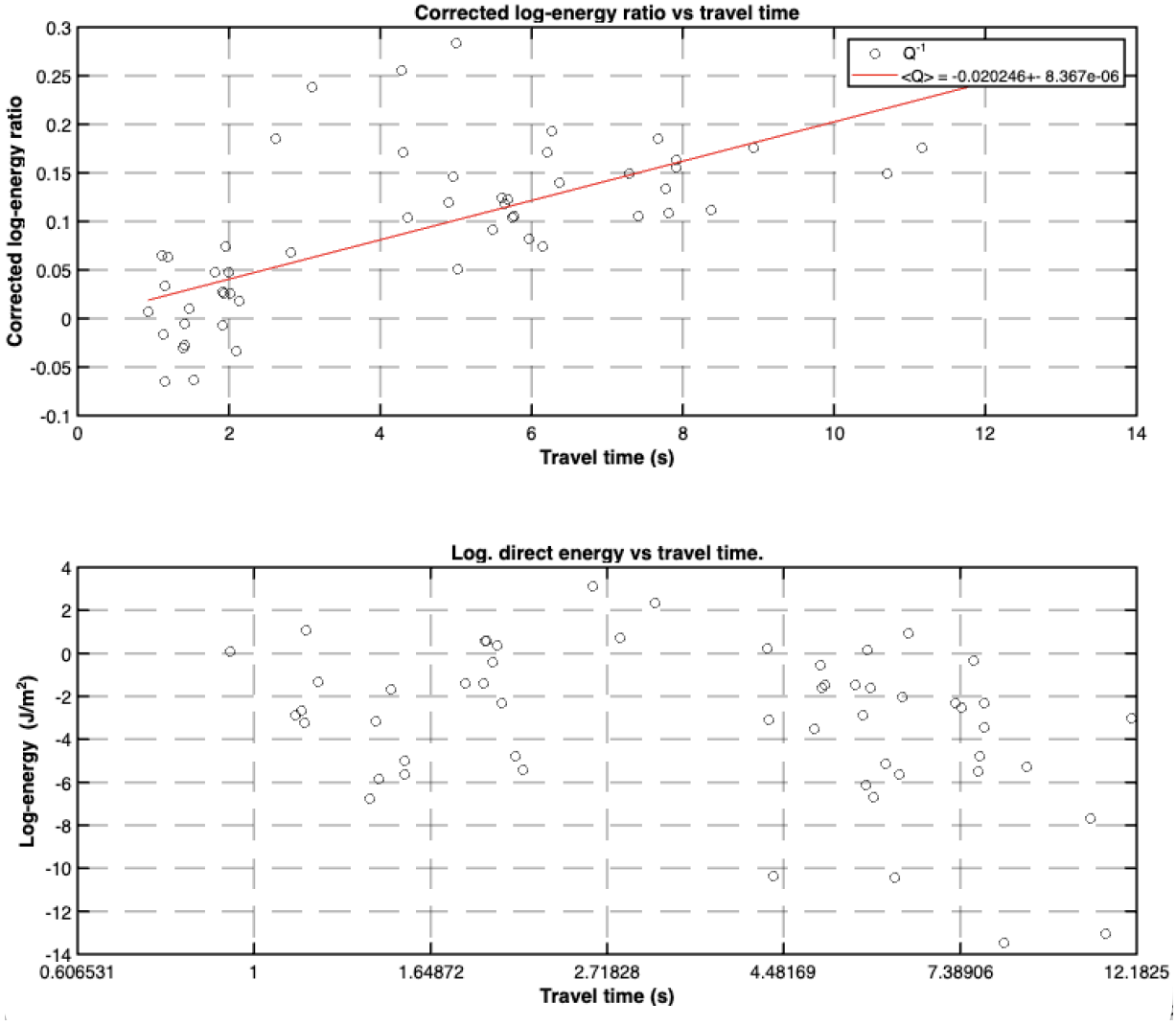


Figure 8.4: Analysis of the data and inversion results for the coda-normalisation inversion at MSH for the 3 Hz frequency band. The upper panel (should) show the decay of the logarithmic energy ratio with travel time for the data (black circle) and the theoretical behaviour with average attenuation. The lower panel shows the dependence of logarithmic energy on hypocentral distance. Note the negative quality factor.

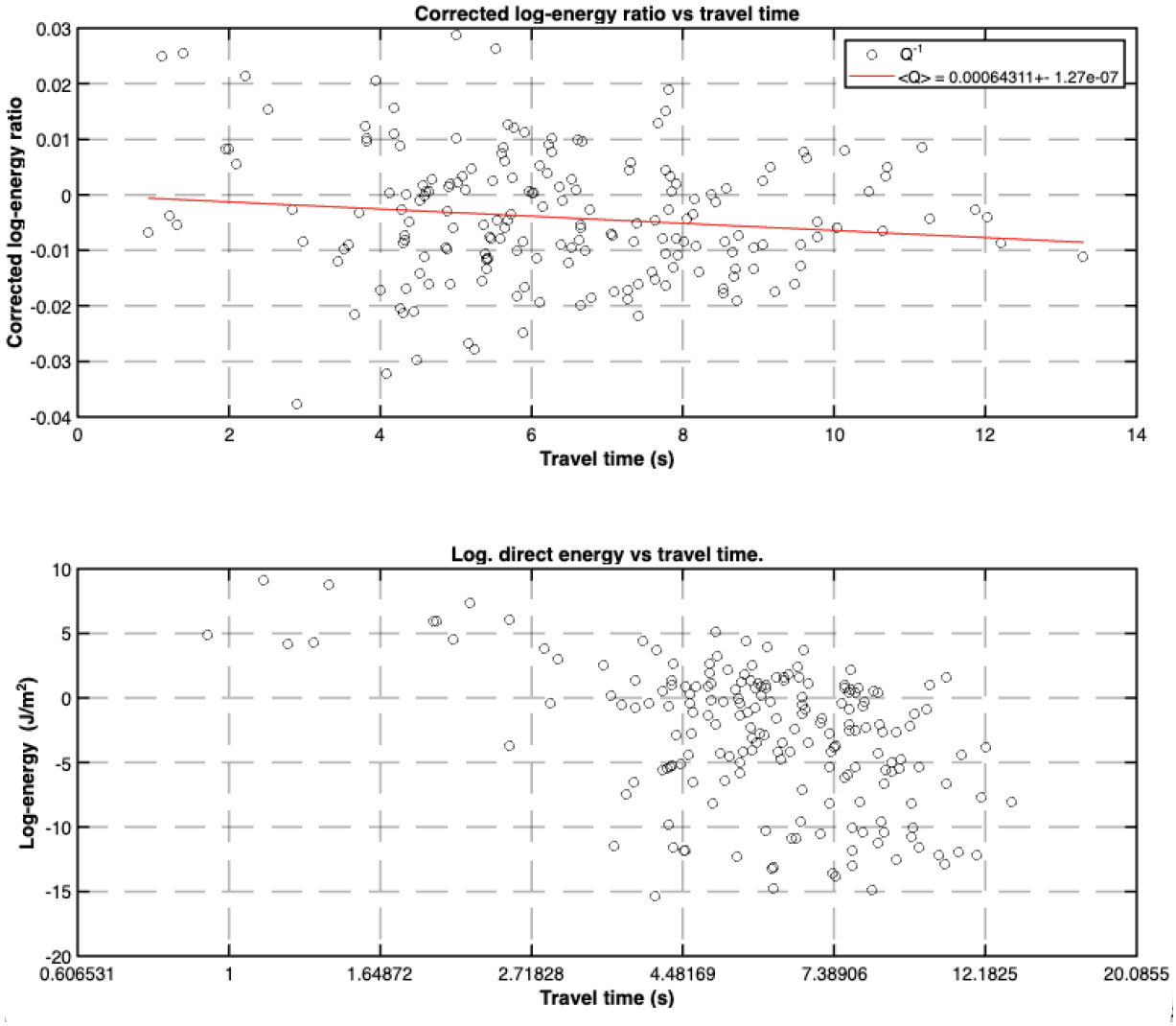


Figure 8.5: Analysis of the data and inversion results for the coda-normalisation inversion at MSH for the 18 Hz frequency band. The upper panel show the decay of the logarithmic energy ratio with travel time for the data (black circle) and the theoretical behaviour with average attenuation. The lower panel shows the dependence of logarithmic energy on hypocentral distance.

Chapter 9

Rays and Kernels

While scattering and total attenuation variations are based on a ray-bending technique (Block, 1991), MuRAT includes 3D sensitivity kernels (Del Pezzo et al., 2018) for modelling seismic absorption in space.

9.1 Ray tracing

MuRAT4 ports the original Fortran code for ray bending from Block (1991) into Matlab. The code is a classical ray bending, starting from the source and receiver locations and bending the ray to minimise travel time inside the input velocity model. The user can look at the function $Murat_tracing$, which *traces* the ray in the velocity model. At the start of the function, two optional parameters set the maximum number of bending iterations and points allowed.

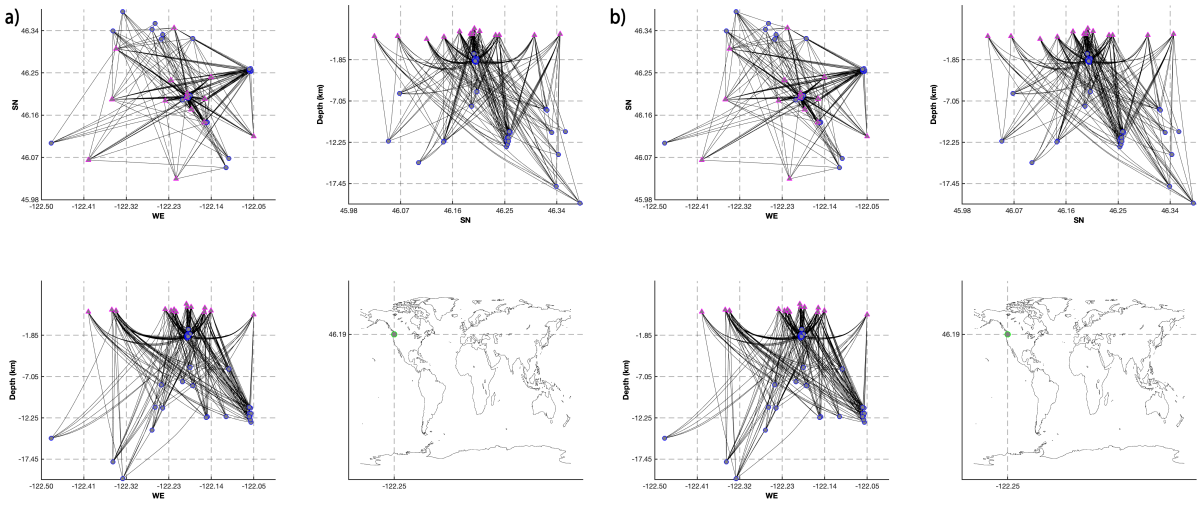


Figure 9.1: Rays of the coda-normalisation (a) and peak delay (b) methods at 3 Hz for the MSH dataset.

Their values (*maxit = 100* and *maxpoints = 10000*, respectively) have been set after several trials and generally work just fine at the crustal scale. The rest of the procedure includes the original function that averages velocities on the regular grid and calculates velocity gradients, informing the bending process (Block, 1991). The functions have been tested against standard Matlab functions and are generally three to ten times faster.

MuRAT pairs to the ray-bending approach some utilities optimised for speed that measure ray lengths across the inversion grid. These lengths are used both in the standard coda-normalisation method and as a weight in the peak-delay regionalisation (Chapters 2-3). The user can look at the function $Murat_rays$, which calls both procedures, to see that the procedure needs the velocity model, a propagation grid, and source/station locations as input. MuRAT calls $Murat_segments$ within the function, calculating segment lengths that it includes directly in the coda-normalisation matrix (see chapter 2) and peak-delay matrix. The corresponding rays are finally shown in the folders **Rays** and **Kernels**. As the data selection is different for coda-normalisation and peak-delay, the code outputs two different sets of **Rays_*** figures (Fig. 9.1). The ray tracing works for both the small volcanic scale (like at MSH, Fig. 9.2a), mantle scale (Fig. 9.2b) and continental-oceanic settings (Fig. 9.2c).

9.2 Kernel implementation

MuRAT implements the computational kernels described by Del Pezzo et al. (2018) in the tomographic procedure described by De Siena et al. (2017a). The kernels (Fig. 9.3) are calculated with a call to the function $Murat_kernels$, using the albedo and extinction lengths input in the code as input. Panel a) shows the entire computational kernel computed up to the laspe time equal to the starting lapse time plus half of the coda window.

MuRAT produces no MLTWA analysis, yet the user needs to set albedo and extinction length. The values in the input files are average for the crust; better measurements are likely available for the area under study.

9.3 Geographic and cartesian coordinates

MuRAT has all inputs in latitude and longitude but needs conversion to cartesian. This conversion is performed mainly in $Murat_checks$ and through other functions within the bin. This approach means we rely on the functions $distance$ and $reckon$ to create, e.g., velocity models for ray propagation or

switch between cartesian and geographic outputs. The implementation in cartesian coordinates is relatively straightforward and already implemented by works at the 10s of meters and laboratory scales - see ***Di Martino et al. 2022*** and ***King et al. 2022***.

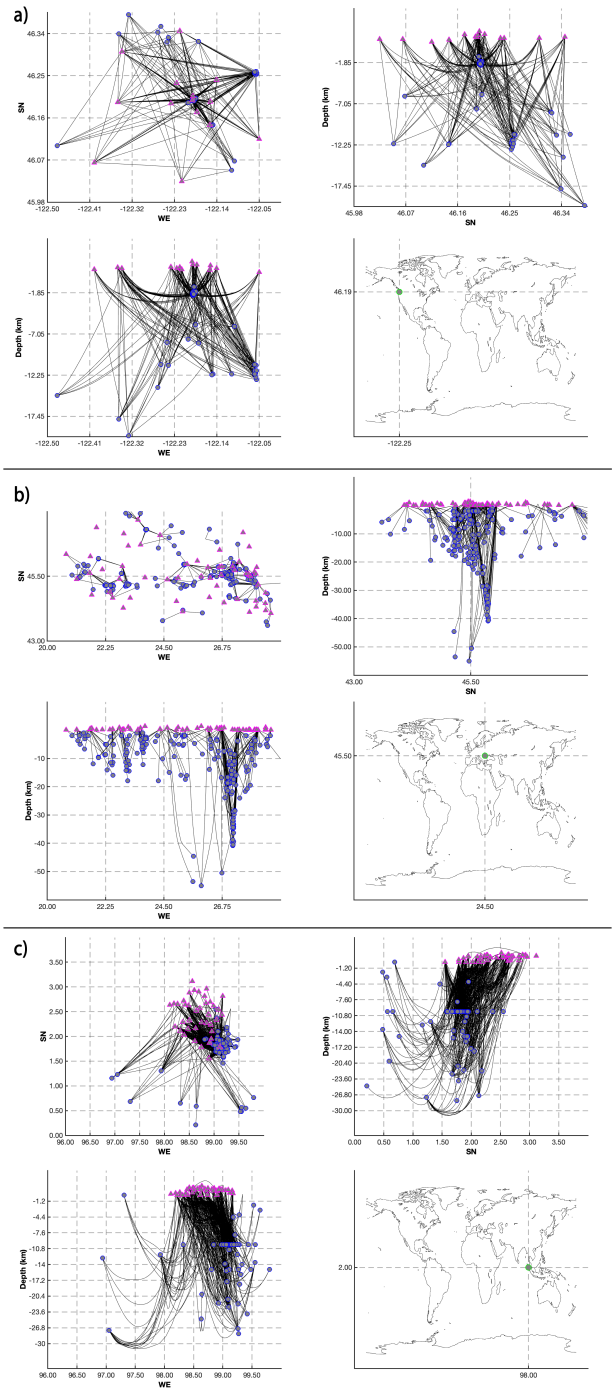


Figure 9.2: Rays of the coda-normalisation method at 3 Hz for the three sample datasets: (a) MSH; (b) Romania; (c) Toba.

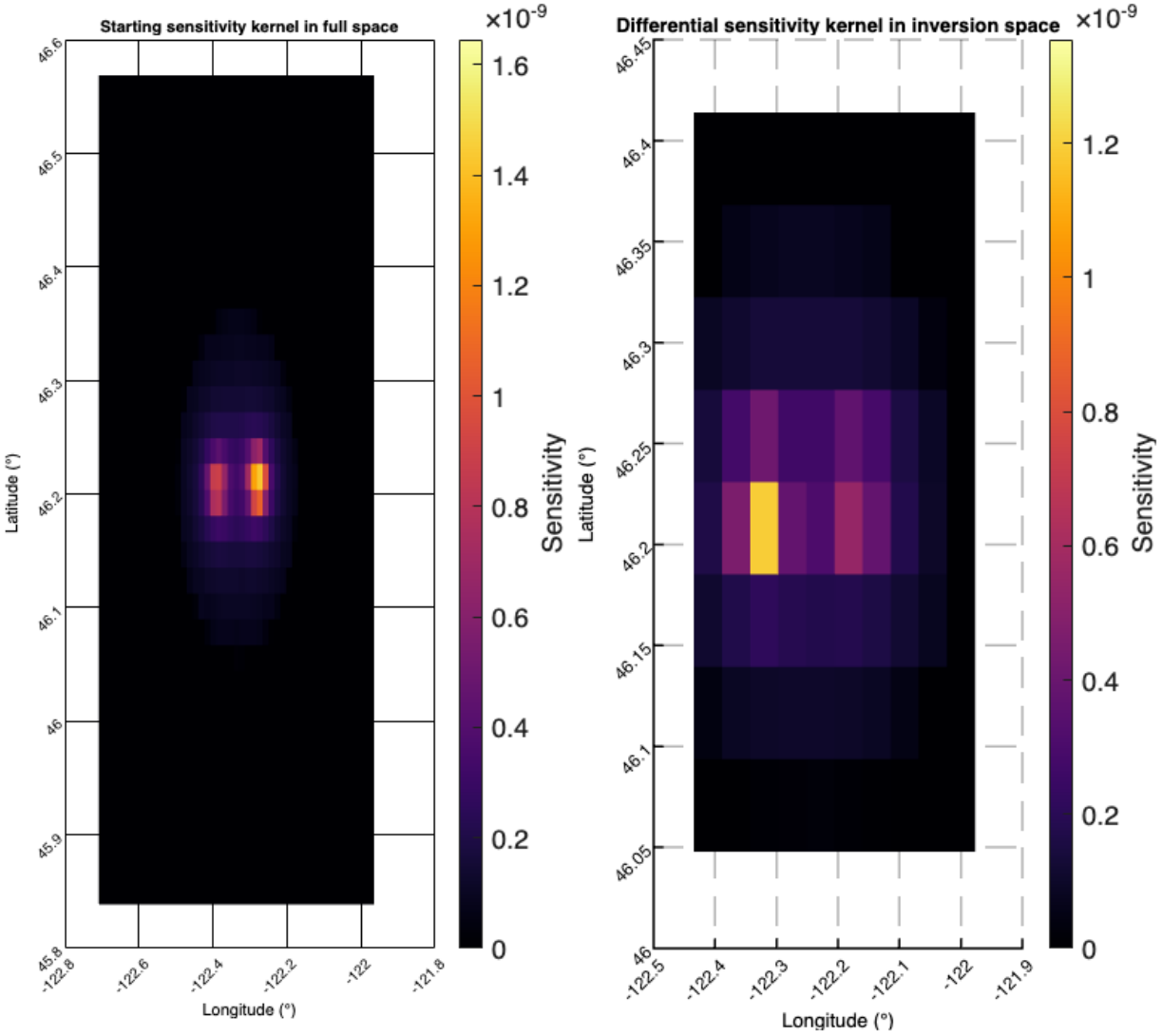


Figure 9.3: Kernels for coda-attenuation imaging at 3 Hz for the MSH dataset. The full kernels are shown as a section in the 3D space (a) and in the grid chosen for the inversion (b). Notice that sensitivity is always highest at the source and receiver. The colour scale is logarithmic. The lapse time is 22.5 s.

Chapter 10

Inversion

The inversion for Q_c and Q requires setting damping and smoothing parameters for each frequency. The L_curve* figures (e.g. Fig. 10.1) show the relation between residual and norm of the solution if using the *regtools* (Hansen, 2007). Nevertheless, MuRAT4 now also employs MATLAB optimization toolboxes for non-linear inversion - we refer to their documentation for further implementation. The code can now output a cost function through time that provides the user a sense of how long it takes to find a global minimum, after setting damping and smoothing parameters. In Fig. 10.2, we show an example for the MSH dataset at 18 Hz when using the particle swarm method. These plots, differently from those for the L-curve, are not saved by MuRAT4. The additional methods require careful setting on maximum number of iterations and stalled iterations.

10.1 Damping and smoothing

For each frequency, MuRAT solves a linear system $\mathbf{Am} = \mathbf{d}$, where $\mathbf{m}$ contains Q_c^{-1} or Q^{-1} in the cells crossed by rays (or with non-zero kernels). For Q_c, data and matrix are weighted by the uncertainty of each measurement. The solution minimises

$$\|\mathbf{Am} - \mathbf{d}\|^2 + (\epsilon\|\mathbf{A}\|)^2\|\mathbf{m}\|^2 + (s\|\mathbf{A}\|)^2\|\mathbf{Lm}\|^2, \qquad \mathbf{m} \geq 0, \tag{10.1}$$

where ϵ is the damping ($Murat.input.dampingValueQc$ or $Murat.input.dampingValueQ$), s the smoothing ($Murat.input.smoothingValueQc$ or $Murat.input.smoothingValueQ$), $\|\mathbf{A}\|$ the largest singular value of the (weighted) matrix, and $\mathbf{L}$ a first-difference operator between adjacent nodes of the model grid. Both regularisation weights are thus **relative to the matrix**: $\epsilon = 1$ weighs model norm and data misfit equally, and the same value acts alike for Q_c and Q, whose matrices can differ by several orders of magnitude, and across datasets. Values between 0.01 and 1 are a sensible range. Left empty, damping is 0.1. Since $\mathbf{L}$ does not penalise a uniform model, smoothing reduces the spread of the solution without pulling it towards zero; if smoothing is set and damping is left empty,

damping is 0.

With the **'Tikhonov'** method, eq. 10.1 is solved with a projected accelerated gradient method (FISTA), starting from the regionalised model. Small damping values (below about 0.1) make the system ill-conditioned: the solver may need more than $Murat.input.MaximumIterations$ iterations and then prints a warning. In this case, increase the iterations or the damping. For **'Particle'**, **'Annealing'** and **'Global'**, the damping enters the cost function in the same relative form, while the smoothing value is the correlation length of a model covariance.

10.2 Damping sweeps

The L-curves show the effect of damping only. To compare models obtained with different damping values directly, set $Murat.input.dampingSweepValues$ to a list of relative values (e.g., [0.03 0.1 0.3]). MuRAT processes the data and computes rays and kernels once, then inverts and plots the results for each value, which is applied to Q_c and Q at all frequencies. Each run is saved in its own folder (Chapter 7). The relative misfit reduction printed for each frequency compares the final misfit with that of the regionalised starting model: values above 1 indicate that damping is too strong for the data to be fitted.

Damping values from input files written before relative damping was introduced are not interchangeable with the current ones and must be chosen again.

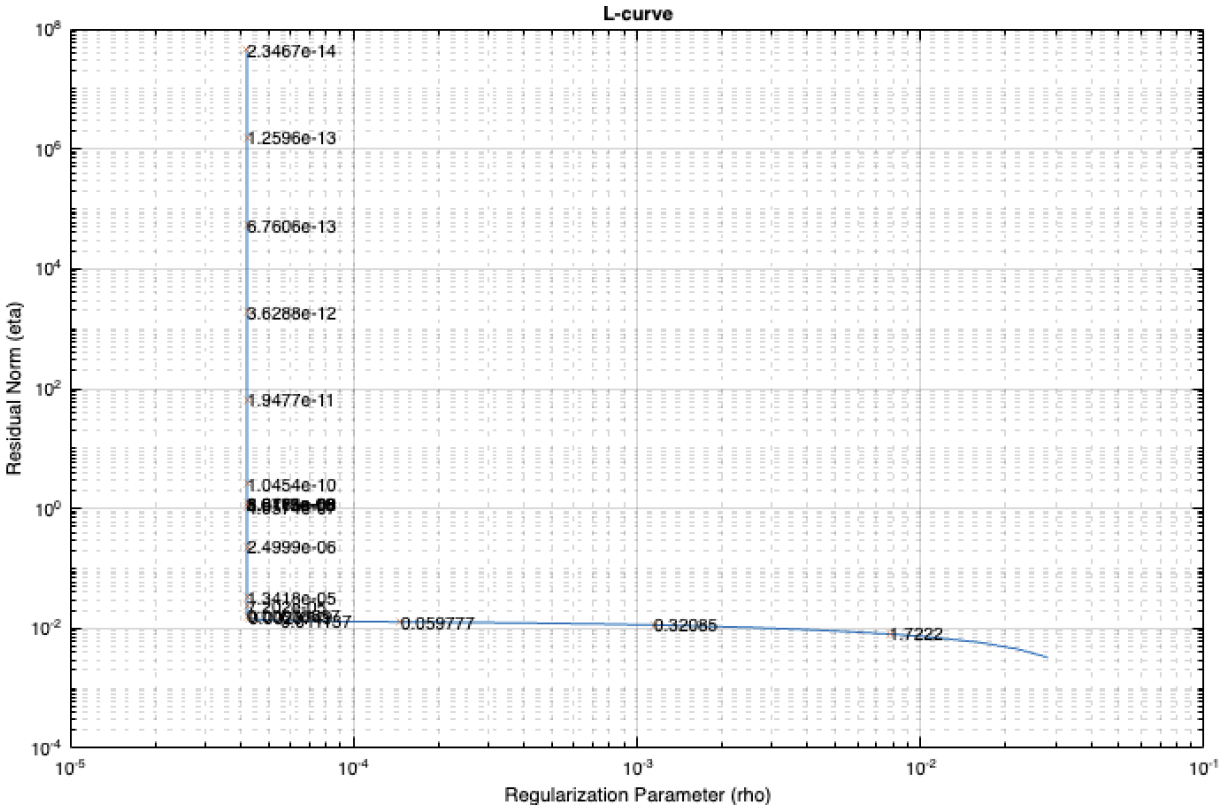


Figure 10.1: Relation between data fit and solution norm for the MSH dataset at 18 Hz.

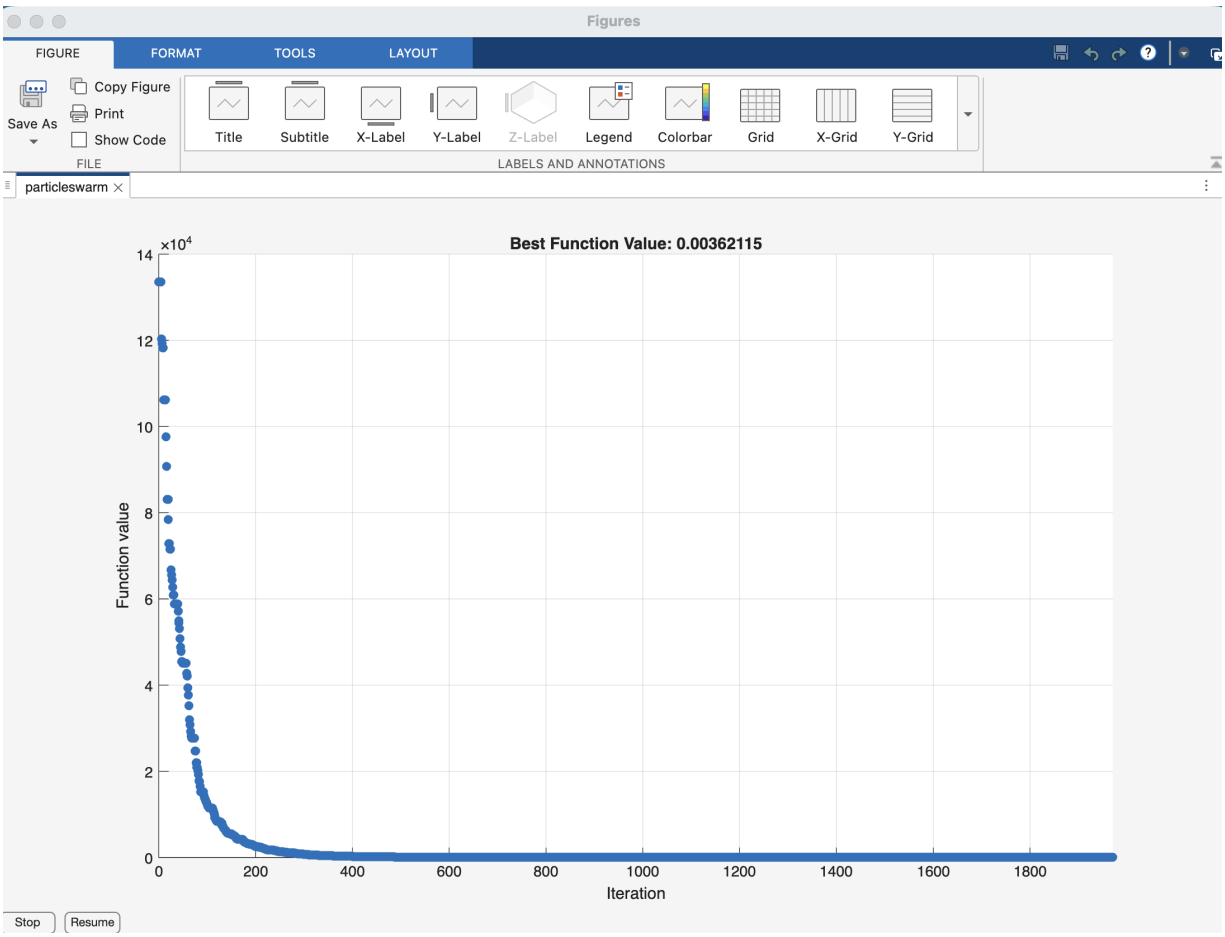


Figure 10.2: Example cost function for the MSH 18 Hz case in the particle swarm inversion.

Appendices

Appendix A

MuRAT input file - Mount St Helens

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INPUT MuRAT - MSH

This is an input file for the program Multi-Resolution Attenuation Tomography (MuRAT), version 3. It refers to the following area:

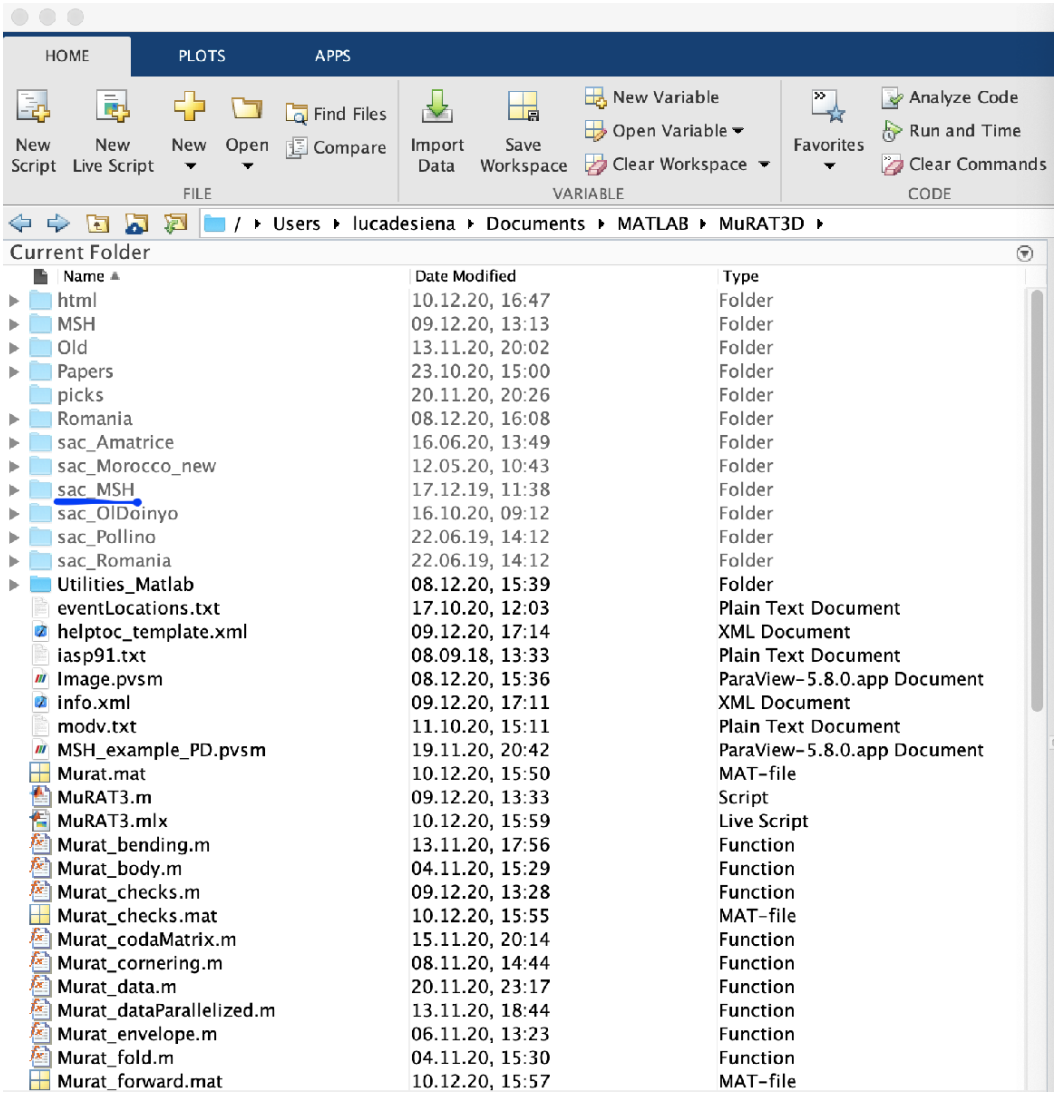
```
MOUNT ST HELENS VOLCANO
```

The working directory is the folder where you downloaded MuRAT: move it anywhere in your system.

GENERAL FIELDS

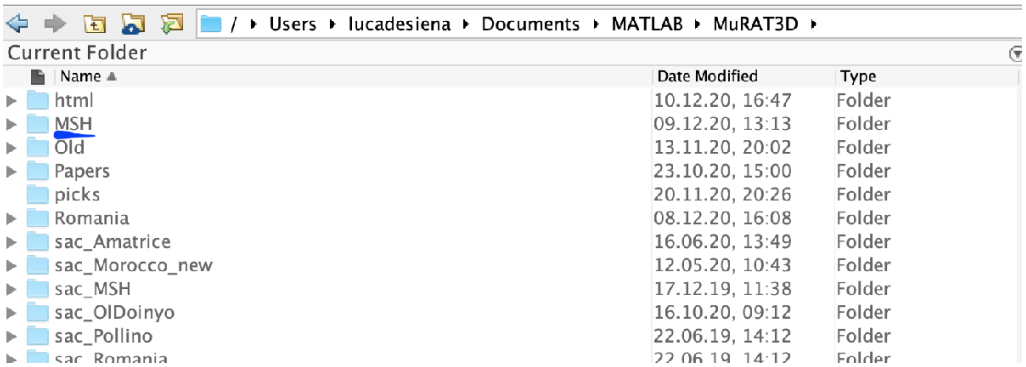
The user needs a data directory containing *.sac* files (beware of the difference between *.sac* and *.SAC*). If you want to use two or three-component recordings, you have to store them into a single folder, but the three components must be in the exact order **E,N,Z**. For an example see Toba. For this example we will use data in the **sac_MSH** folder. Inside this folder, only vertical (*Z*) seismograms are stored:

```
Murat.input.dataDirectory           =   'sac_MSH';
```



Best practice is to create a new folder to store your results. The code will create this folder inside the working directory. Specify the name of the folder that will store text files and figures, and will appear in your working directory:

```
Murat.input.label                    =    'MSH';
```



In MuRAT3D you can choose between a sequential or parallelized forward loop. In the parallelized case, just set a number of workers (cores). In this example, we are working with the parallelized code and a computer with 8 cores. For a sequential run, set *Murat.input.workers* as empty (**[]**).

```
Murat.input.workers                 =  8;
```

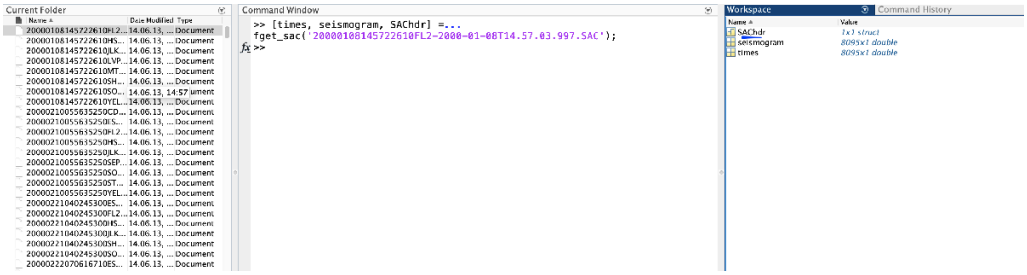
WAVEFORM DATA

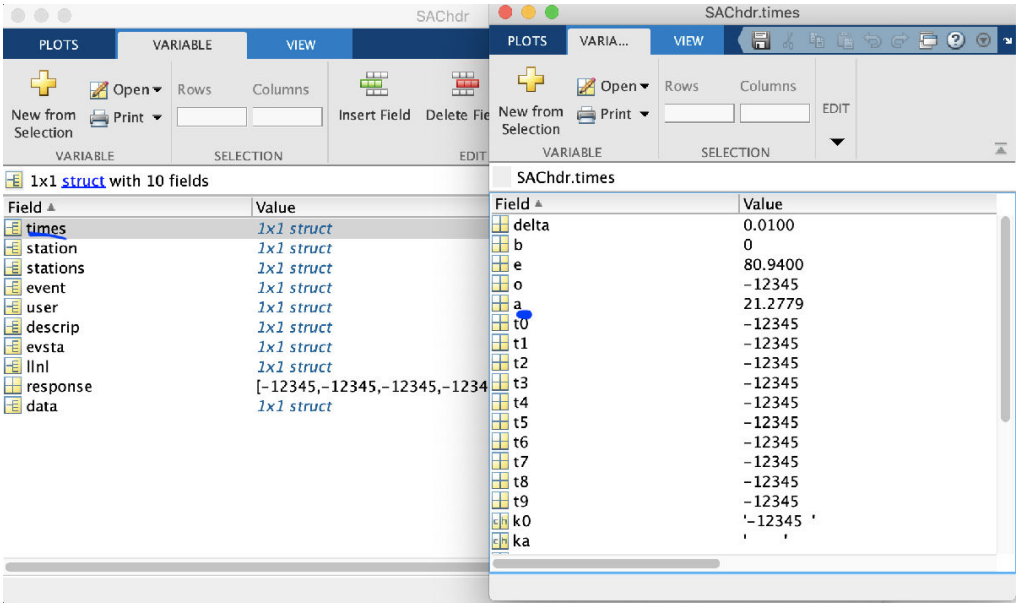
Set all the variables required by data processing. This includes data choises, as setting the name of the variables in SAC and all the attributes that are needed for the three kinds of imaging. The routines for loading the files have been mostly created by Zhigang Peng and co-workers and downloaded from their Introduction to SAC webpage.

MuRAT3D is designed to work with *sac* files only, so it is necessary to set the name of the variable containing their pickings and zero time. Files have to have populated headers. Set the name of the variable containing the origin time, P-wave pickings and S-wave pickings - in this example there is only the mandatory P-wave picking. If you don't have the origin time and S-wave times at hand you can mark them as empty (**[]**):

```
Murat.input.originTime              =   [];
Murat.input.PTime                   =   'a';
Murat.input.STime                   =    [];
```

SAChdr is a structure that contains all the fields you saved in the SAC haeder. In this example the P-wave picking has been stored inside variable *a.* You always find these variables in the *times* subfield:





Then choose the coherent phase you are analyzing - P-(**2**) or S-(**3**). In our case it is P- as we have no S-wave picking:

```
Murat.input.POrS                    =    2;
```

You need to set the central frequencies (Hz) and envelope smoothing time (s) according to your spectrograms and waveform data. General practice is to vary it across your spectra (see De Siena et al. 2016, EPSL) for absorption and scattering mapping or focus on a given frequency (De Siena et al. 2014, JGR) for direct-wave attenuation imaging. Here, they cover the interval [3-18] Hz.

```
Murat.input.centralFrequency        =    [3 6 12 18];
```

Envelope smoothing time to improve fit with equation. If not specified, a default value of 1.0 s is used.

```
Murat.input.envelopeSmoothTime      =   1;
```

You can work with 1 vertical or horizontal (*1*), 2 horizontal (*2*) or three components(*3*). If using more than one component, the order *MUST BE: WE, SN, Vertical or SN, WE,* Vertical. In this example we work with the vertical component only:

```
Murat.input.components              =    1;
```

Sometimes you want to exclude waveforms depending on minimum and maximum travel times, as these data do not fullfill the theory (you generally see this after running the tests). In this case you can set minimum and maximum travel times, so that you exclude traces outside of these intervals. Otherwise, you can set the minimum to **0** and the maximum to **Inf**.

```
Murat.input.mintravel               =   0.2;
Murat.input.maxtravel               =   13;
```

Finally, you can opt to decluster your data events. The code will divide the inversion grid by the following factor and select the best earthquake located in the block among all others. Set it to empty if you want to opt out (**[]**).

```
Murat.input.declustering            =    5;
```

PEAK DELAY

Scattering (peak delay) measurements rely on the existence of coherent waves. Peak delay is a standard measurements of forward scattering in regional scale-mapping since Takahashi et al. 2007, GJI. Here, we approach the problem by adding a ray-tracing strategy to the system, assuming that the sensitivity follows the seismic ray. Also, we bring it to 3D and apply it to P-waves (see De Siena et al. 2016, EPSL).

To do so we need to set a minimum peak delay (*s*) considering scattering in the area and frequency. If using P wave this is important, as they can be sometimes more energetic than S-waves, biasing our measurements. We also need to set the maximum peack-delay (*s*) to avoid picking surface waves.

```
Murat.input.minimumPeakDelay        =    0.1;
Murat.input.maximumPeakDelay        =    7;
```

The user has the option to keep only those peak delays whose logarithmic trend is nearest to the average trend obtained by removing geometrical spreading. See Napolitano et al. 2020 GF and Gabrielli et al. 2023, GRL, for details. Set the following variables to the amount of standard deviations from the average trend you want to consider - the selection will be apparent in the Peak Delay test results. Set high numbers to consider all your data.

```
Murat.input.stDevPD                 =   2;
```

DIRECT WAVE ATTENUATION

Total attenuation (inverse Q) measurements also relies on the existence of coherent waves. Total attenuation with the coda-normalization method is today a standard in volcano tomography (Del Pezzo et al. 2006, PEPI, De Siena et al. 2010, JGR; De Siena et al. 2014, JGR, Prudencio et al. 2015, Prudencio & Manga, 2020.

The method relies on the measurements of both direct and coda wave energy. Then we set the length of the window used to measure direct-wave energy, trying to smooth radiation pattern effects. The code uses the same length to measure noise energy. Discussions on the topic can be found in De Siena et al. 2009, PEPI and De Siena et al. 2010, JGR, but the choise is as usual dependent on data:

```
Murat.input.bodyWindow              =    1;
```

We also need the start of the window used to measure noise a few seconds after the start of the recording (in seconds).

```
Murat.input.startNoise              =    5;
```

The coda-to-noise energy ratio is used by the weighted inversion. Here you set the minimum accepted energy ratio.

```
Murat.input.tresholdNoise           =    3;
```

CODA ATTENUATION

Coda attenuation (inverse Qc) is measured from the decay of the coda with lapse time from the origin time of the earthquake, and requires energetic scattering. Qc is a well know parameter for assesing tectonic structures (Sato et al. 2012, Springer). In recent years it has been used as an imaging attribute at the regional (Calvet et al. 2013, Tectonophysics; Borleanu et al. 2017, Tectonophysics), fault (Napolitano et al. 2020; Sketsiou et al. 2020, PEPI) and, especially, volcanic scales (Prudencio et al. 2013a, GJI; Prudencio et al. 2013b GJI; De Siena et al. 2016, EPSL; De Siena et al. 2017, GRL; Gabrielli et al. 2020, GJI).

Lapse times and sensitivity kernels

In MuRAT3D, we use the full 3D computational kernels devised by Del Pezzo et al. 2018, Geosciences in the inversion approach proposed first by De Siena et al. 2017, GRL.

Here you choose the start of the window used to measure coda wave energy and model kernels. The starting time of the coda window can be set directly in seconds ('**Constant**'), from the envelope peak ('**Peak**'), or depending on the P- or S-wave travel time, for example twice the S-wave travel time ('**Travel**'). If the chosen method is **Constant**, set start and length of the window.

```
Murat.input.lapseTimeMethod         =   'Constant';
```

If the chosen method is **Constant**, set the start of the window in seconds after the origin time. If it is **Peak**, set it to **[]**. If it is **Travel** select *Murat.input.startLapseTime* as the moltiplicative factor of the phase you are using (e.g., **2** or **3**). Avoid this method if you do not know the S-wave time.

```
Murat.input.startLapseTime          =   15;
```

Finally set the length of the coda window in seconds. The true lapse time at which we calculate the kernels is half of the window. The window is also used (after normalizing for its length) in the coda normalization method.

```
Murat.input.codaWindow              =   10;
```

The **MLTWA** is the standard method to find the average parameters necessary to calculate the kernels. It provides albedo and inverse extinction length:

```
Murat.input.albedo                  =   [0.8 0.6 0.4 0.2];
Murat.input.iExtinctionLength       =   [0.12 0.12 0.1 0.1]; % D  =   vS./
Le_1/3./(1-B0)
```

As the kernels are computational intensive and require a full matrix of nodes to avoid singularities, we also use a computational factor to reduce the computational time. This number divides the input grid, meaning that higher numbers give more precise results at the expene of computational time. Minimum is **1**. A figure will output the kernel in this grid.

```
Murat.input.kernelTreshold          =    2;
```

Measurement of Qc

MuRAT3D implements either a linearised approach or a grid search approach to measure Qc. The linearised approach is the standard proposed first by Aki (e.g., Havskov et al. 2016, BSSA) to best fit Qc after taking the logarithm of the energy. The uncertainties are derived from the simple minimum R-squared (fitTresholdLinear) and needs to be defined by a number between 0 and 1. It is advisable to set a minimum of 0.1.

The non linear approach models energy data measured on one-second windows across the envelope and minimizes the difference between data and model with a 1D grid search algorithm (Napolitano et al. 2020). Uncertainties are given by the experimental probability density function of the misfit. In both cases, uncertainties play as a weight in the final inversion. In the second case, leave the fitTresholdLinear = **[]**.

The user needs to choose between the two options **'Linearized'** and **'NonLinear':**

```
Murat.input.QcMeasurement           =   'NonLinear';
Murat.input.fitTresholdLinear       =    0.006;
```

GEOMETRY AND VELOCITY

This section sets the details of the inversion grid and availability of velocity model. In MuRAT3D the coordinates of the model are in lat/lon, then they get converted in km. The vertical is in altitude above sea level. The velocity model can be 1D or 3D - if 3D all points must be given in lat/long formats. You start by setting the origin and end points of your inversion grid.

```
Murat.input.origin                  =   [45.9805 -122.5030 3350];
Murat.input.end                     =   [46.3900 -122.0000 -22650];
```

Then you need to set the number of nodes in the three directions. This is obviously dependent on the scale of your area. You will be playing a lot with this to test your effective resolution capabilities.

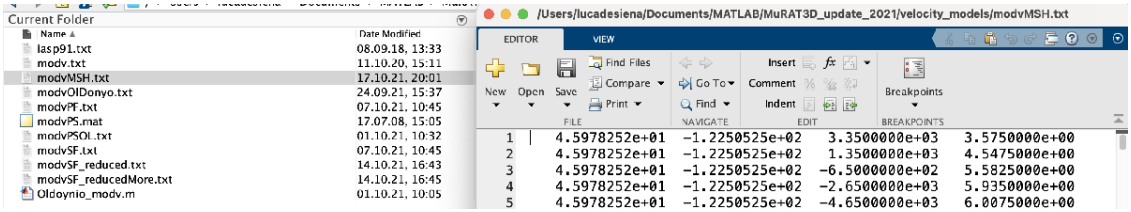
```
Murat.input.gridLat                 =   10;
Murat.input.gridLong                =   12;
Murat.input.gridZ                   =   6;
```

You will see all of your figures on three sections cutting the models WE (degrees), SN (degrees), and horizontally (meters or km) at:

```
Murat.input.sections                =    [46.12 -122.20 -1000];
```

With this version of the code you are always using an underlying velocity model: the 3D is either unavailable (***0***) or a available (***1***) velocity model. For the 1D case MuRAT provides both the *iasp91.txt*, the standard IASPEI velocity model and the more accurate Lithos model, and expands any of them to a false 3D. However, a standard crustal model is generally available everywhere on the Earth, so use that, but change it to the same format as the file provided: first column is depth, second is distance from the centre of the Earth, then third and fourth are P- and S-wave velocity. Store the file in the folder **velocity_models**. See the Romania example file.

At Mount St. Helens, we use the 3D local earthquake tomography model of Waite and Moran, 2009,JVGR, (*modvMSH.txt*), stored in the folder **velocity_model:**



whose coordinates are Lat/Long/Altitude/Velocity (P or S depending on the phase you are using) in meters. So we set:

```
Murat.input.availableVelocity       =    1;
Murat.input.namev                   =   'modvMSH.txt';
```

Even if we set the velocity model we still need the average crustal velocities if you have no info of origin time. It is highly recommended you have the origin in the haeder, at variables **'o'**!

```
Murat.input.averageVelocityP        =   6;
Murat.input.averageVelocityS        =   2.5;
```

INVERSION

The code implements:

- a standard Tikhonov inversion based on singular value decomposition (**'Tikhonov'**), where we on the regtools Matlab suite from Per Christian Hansen.
- the particle swarm (**Particle**), simulated annealing (**Annealing**), or genetic algorithm (**Genetic**) solvers from the optimization toolbox.

In the last case, the user can choose the maximum number of iterations after which the optimization will stop. The user can also choose the maximum number of iterations where the misfit stalls after which the optimization will stop. Otherwise leave empty.

```
Murat.input.inversionMethod         =   'Particle';
Murat.input.MaximumIterations       =   2e3;
Murat.input.MaximumStallIterations  =   100;
```

Set this to ***1*** to plot during computation:

- the L curves between residual and norm length (Aster et al. 2013 - case **'Tikhonov'**).
- the decrease in misfit for the remaining cases.

```
Murat.input.PlotInversion           =   0;
```

The user can set damping and smoothing parameters for Qc. The first will increase the impact of the norm of the solution on the objective function. The damping is set to 0.1 of the average inverse coda quality factor if left empty, otherwise the user sets multiples of this value. The smoothing is set to three times the step of the grid in the vertical direction, otherwise the user sets multiples of this value.

```
Murat.input.dampingValueQc          =   [5000 1 1 0.1];
```

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